

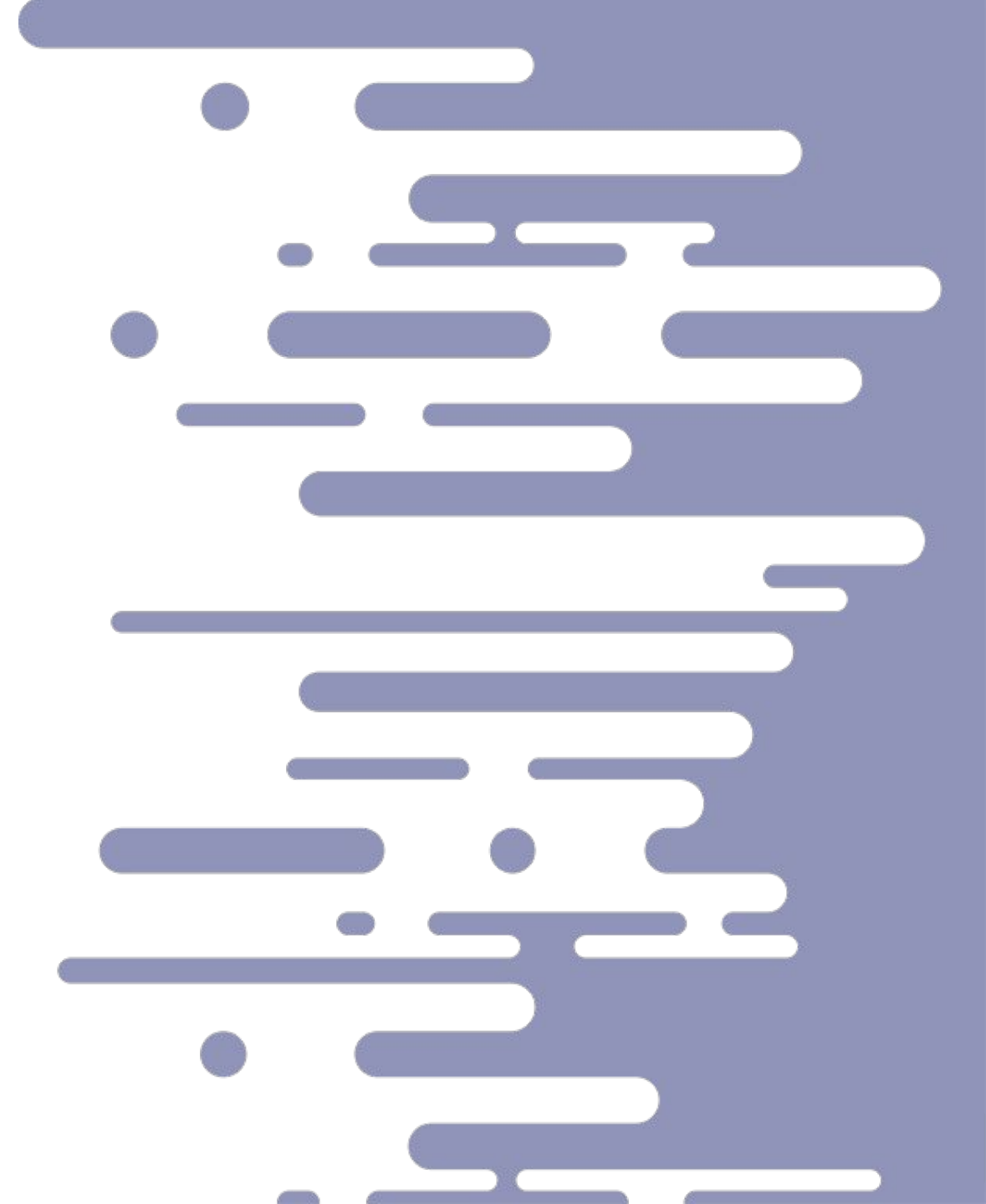
Sylvie Campagne



European Institute
for Energy Research
by EDF and KIT

Introduction to systematic screening

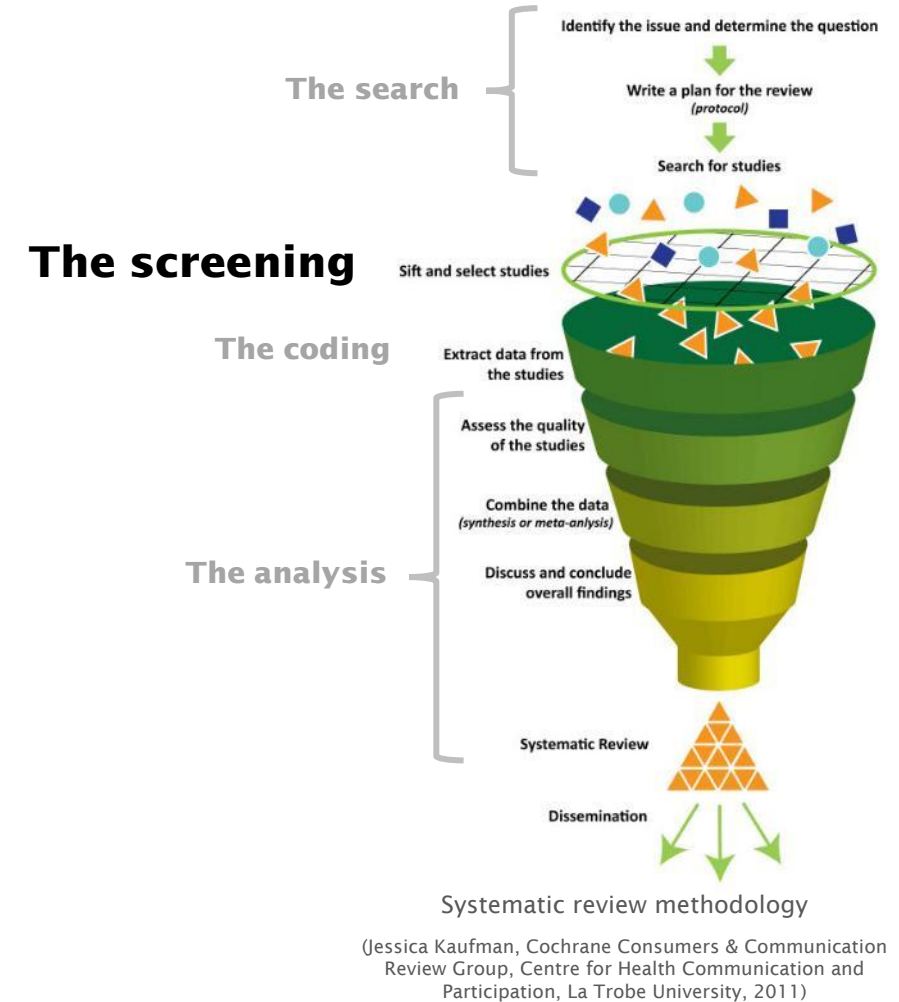
Formation FRB-CESAB 6th to 10th October 2025



The screening phase

The screening phase

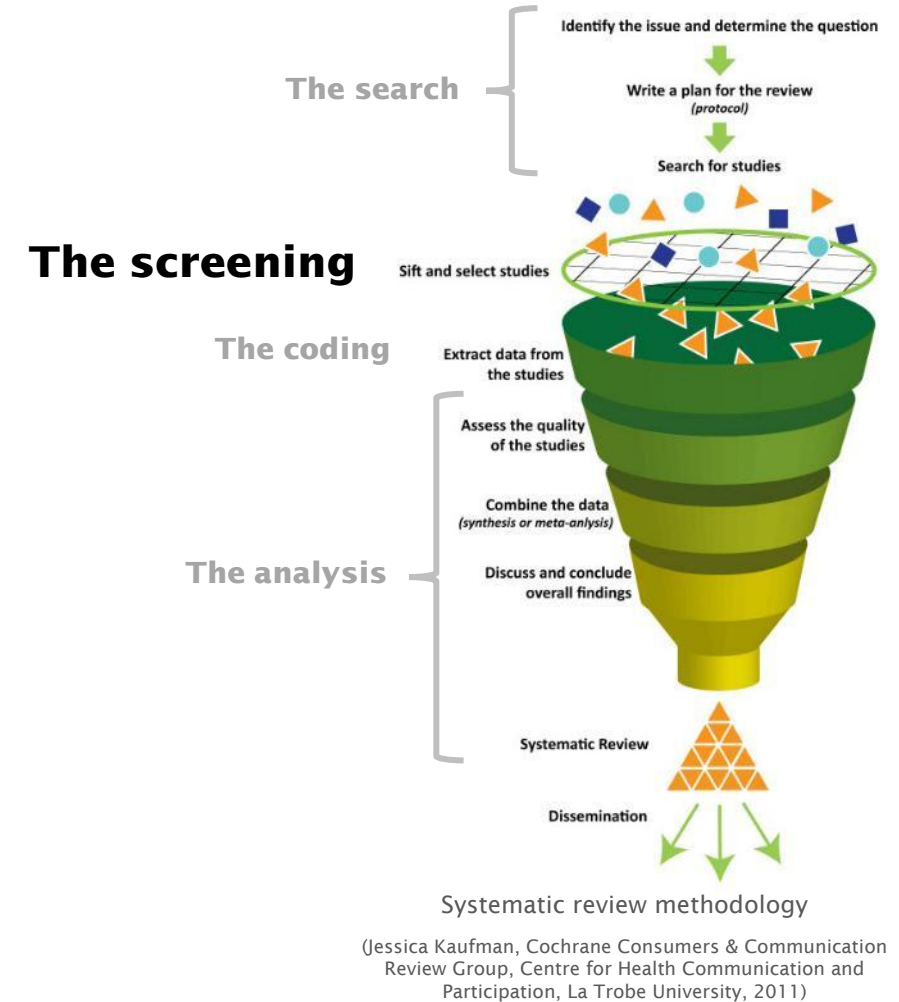
Select the references related to our topic or question from the references found during the research phase.



The screening phase

Select the references related to our topic or question from the references found during the research phase

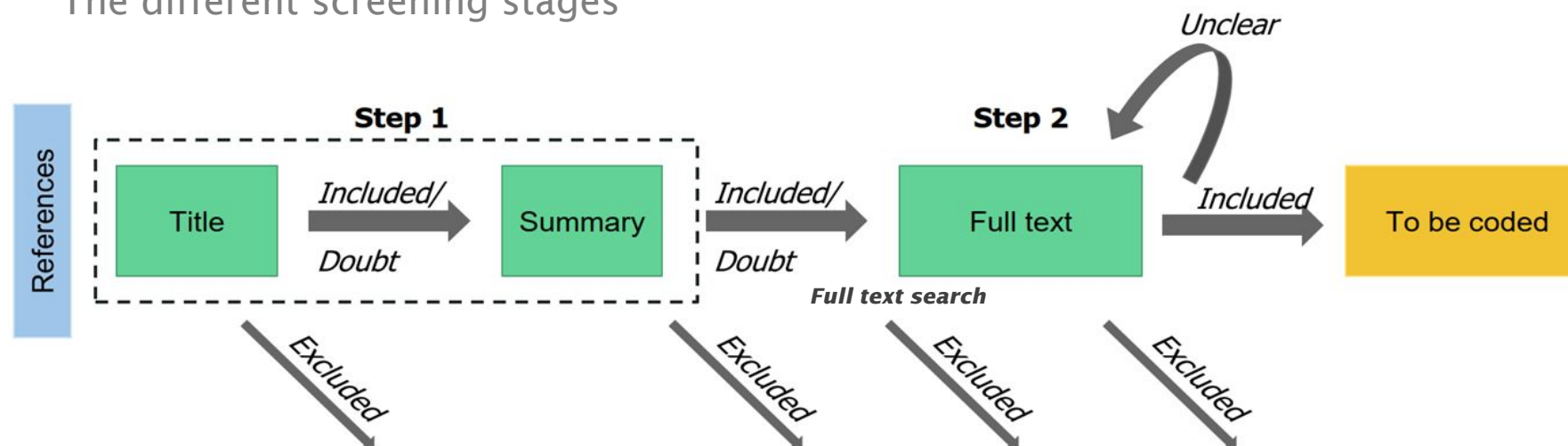
1. The different screening stages
 2. Inclusion and exclusion criteria
 3. Existing tools for managing screening
 4. Statistical tests between raters (kappa test)
- Check for duplicates
 - Title / Abstract screening
 - Full text searching
 - Full text screening



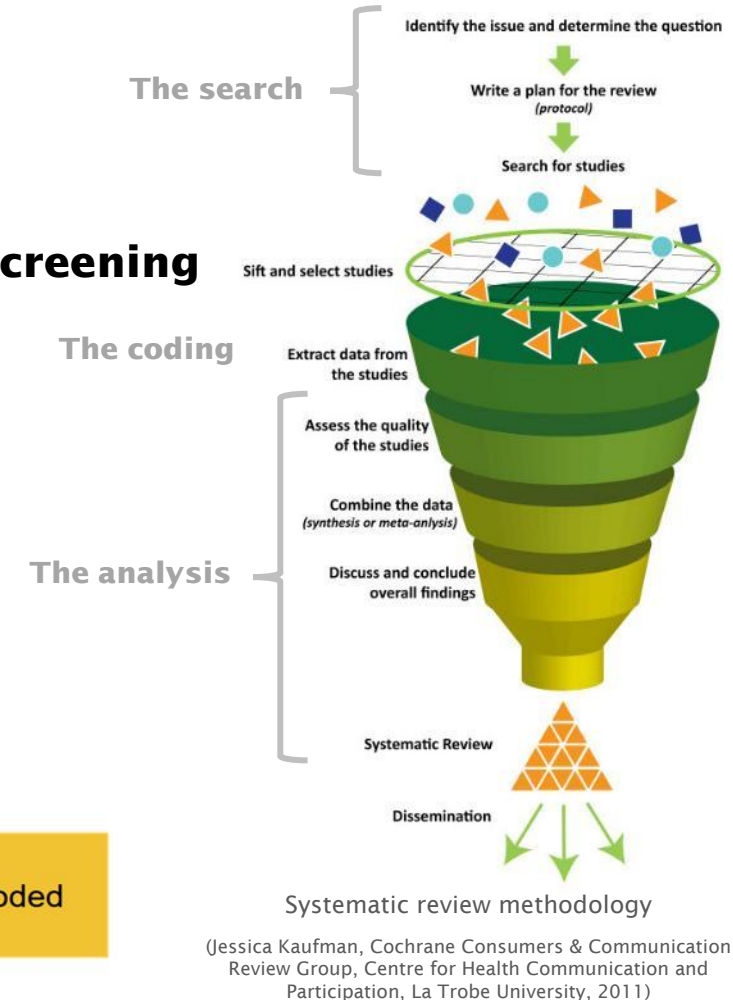
The screening phase

Select the references related to our topic or question from the references found during the research phase

1. The different screening stages



The screening



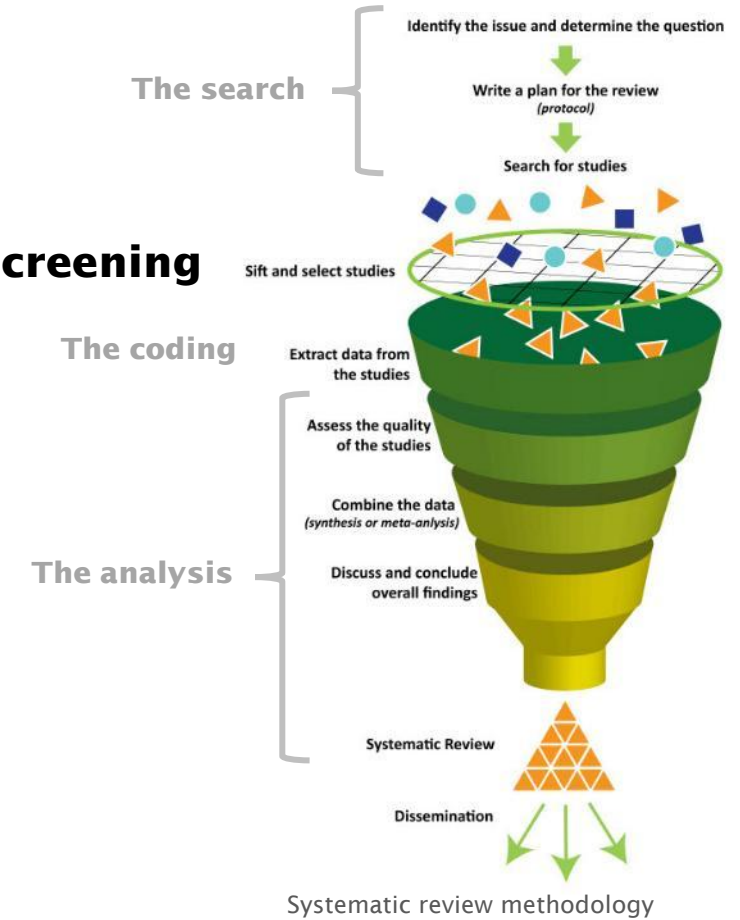
The screening phase

Select the references related to our topic or question from the references found during the research phase

- Title / Abstract screening
- Full text screening

Inclusion and exclusion criteria

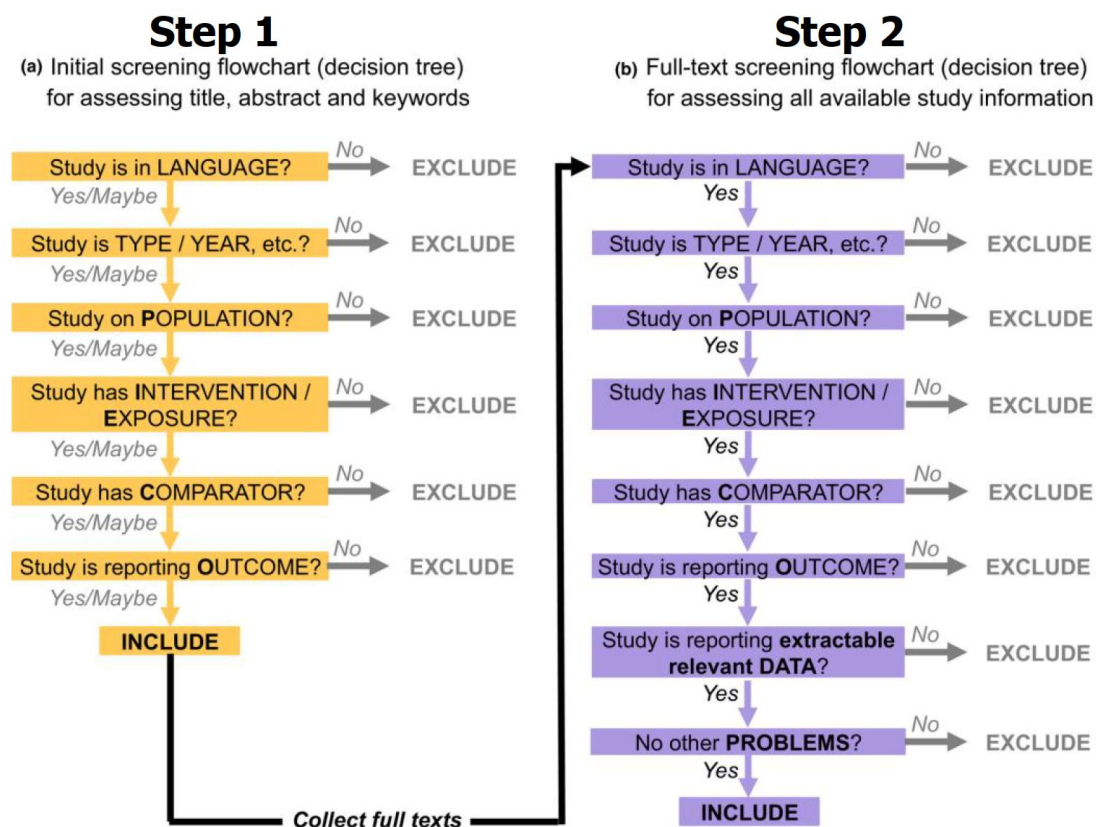
The screening



(Jessica Kaufman, Cochrane Consumers & Communication Review Group, Centre for Health Communication and Participation, La Trobe University, 2011)

The screening phase

Inclusion and exclusion criteria



Decision trees in the screening phase

(Foo et al, 2021)

The screening

The coding

Ideally, at each stage :

→ generate the decision tree using **PICO** and **IN/OUT** criteria

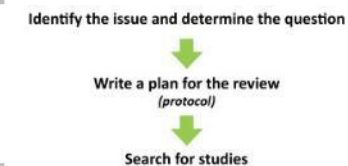
→ **discuss** the decision tree (with at least 1 other evaluator)

→ **benchmark** the decision tree (on a few articles, 2+ reviewers, compare results)

→ **refine** the decision tree

SORTING

The search



Sift and select studies



Extract data from the studies

Assess the quality of the studies

Combine the data (synthesis or meta-analysis)

Discuss and conclude overall findings

Systematic Review

Dissemination

Systematic review methodology

Patricia Kaufman, Cochrane Consumers & Communication Review Group, Centre for Health Communication and Participation, La Trobe University, 2011)

The screening phase

Inclusion and exclusion criteria

Decision trees (Campagne et al, 2023)

Criterion	Screening step	Inclusion criteria	Exclusion criteria
<i>Population</i>	Title	Articles whose title deals with biodiversity, i.e., species, habitats, and/or ecosystems in marine environments. Non-exhaustive examples may include open-ocean, continental shelf, coastal areas, seagrass meadows, estuaries, mangroves, coral reefs, etc.	Articles whose title explicitly only refers to terrestrial and/or freshwater biodiversity, species, habitats or ecosystems, i.e., articles regarding exclusively aquatic species and habitats (e.g., lakes, floodplains, rivers, subterranean habitats, etc.) or to terrestrial species and habitats (e.g., forest, agricultural ecosystems, etc.)
<i>Outcomes</i>	Title	Articles dealing with marine ecosystem services (as well as related terms such as “nature’s contributions to people”). (e.g., marine blue sequestration, snorkelling, whale watching) Articles dealing with the marine ecosystem service of food supply in terms of indicators of stock or population size of commercial species (e.g., fishery stock)	Articles dealing solely with function or structure processes and not related to effects on ecosystem services (e.g., primary production, photosynthesis) Studies only addressing species criteria with indicators other than the stock or the population size of the species (e.g., species distribution)
<i>Exposure</i>	Abstract	Any article or study exposing marine biodiversity, i.e., species, habitats, and ecosystems, to a change in structure and/functioning over time caused by an agent of change, i.e., human activity (e.g., direct/overexploitation, land/sea use change, etc.) or a change caused by different spatial area studied	Articles presenting no exposure to a change
<i>Comparator</i>	Abstract	Articles studying changes in ecosystem services through time or space (i.e., temporal or spatial comparisons). This may mean a different study type as detailed in Table 4. Accepted with synchronic comparators (same time, different sites).	Articles only assessing ecosystem services at one time or in one site/area
<i>Temporal period</i>	Abstract	Articles analysing relevant outcomes with data covering periods of at least part of the 20 th century and/or the 21 st century	Articles analysing data covering periods ending before 1900 (e.g., palaeoecology analysis).
<i>Outcomes</i>	Full text	Articles analysing relevant outcomes containing qualitative or quantitative values of marine ecosystem services and disservices	Articles without qualitative or quantitative values of marine ecosystem services and disservices (e.g., narrative review, opinion paper, policy paper without new quantitative or qualitative values defined).

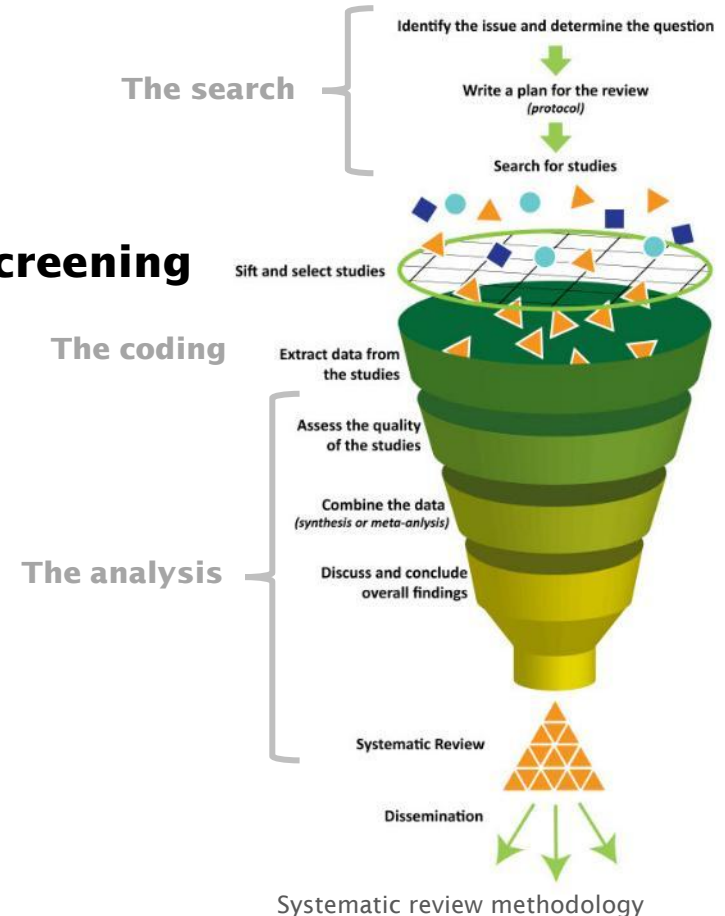
The screening phase

Inclusion and exclusion criteria

Screening steps	Criteria	Inclusion criteria	Exclusion criteria
Title	Ecosystem service	Study assessing any ecosystem service (as well as related terms of ES like “nature contributions to people”) no matter the types of values	Study on the ecological structure, process or function of the desert (e.g. article on primary production) and/or without ES values
Title	Type of ecosystems	Values on wetlands and related ecosystems	Study not on wetlands or without ES results/values on wetlands
Abstract Full-text	Type of articles	Scientific and technical articles and reports; doctoral theses	Methodological papers

- ✓ Increasingly precise criteria at each stage of sorting while maintaining previous criteria
- ✓ A priori criteria preserve transparency and repeatability and minimize bias.
- ✓ When uncertain, be inclusive
- ✓ Decisions to be made according to different situations and must be transcribed for transparency and repeatability
- ✓ There may be criteria not related to PECO, on the language of the article, the type of articles (eg review), the quality or the type of data

The screening



(Jessica Kaufman, Cochrane Consumers & Communication Review Group, Centre for Health Communication and Participation, La Trobe University, 2011)

How to do a systematic review?

The software

Tips for efficient sorting

- Report the exclusion decision
- Search the library for relevant keywords - filter these articles to sort them together
- Work in blocks of 30-45 minutes
- Work simultaneously with other people (facilitates quick consultation)
- BUT BE CAREFUL of any exclusion without a human reading the article!

Excel Microsoft / WPI / Office - free

Need to be very organized - difficulty when evaluating with multiple reviewers.

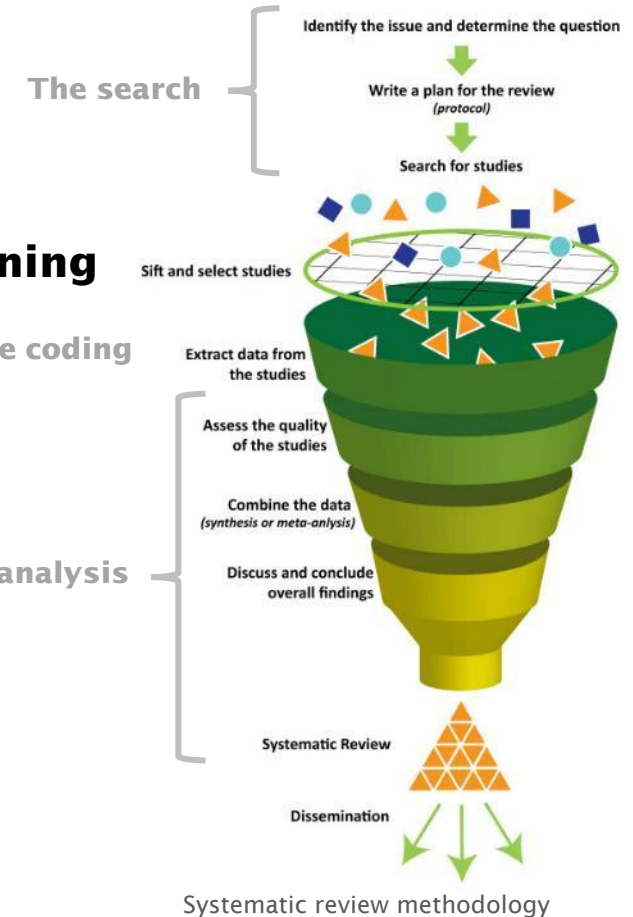


No. of articles	Article title	Sort by title	Abstract sorting	Pdf found	Sort entire text
23	Evaluation of	Yes	NO	-	-
24	Ecosystem...	NO	-	-	-
2X	Mapping...	Yes	Yes	Yes	No

The screening

The coding

The analysis



Systematic review methodology
(Jessica Kaufman, Cochrane Consumers & Communication Review Group, Centre for Health Communication and Participation, La Trobe University, 2011)

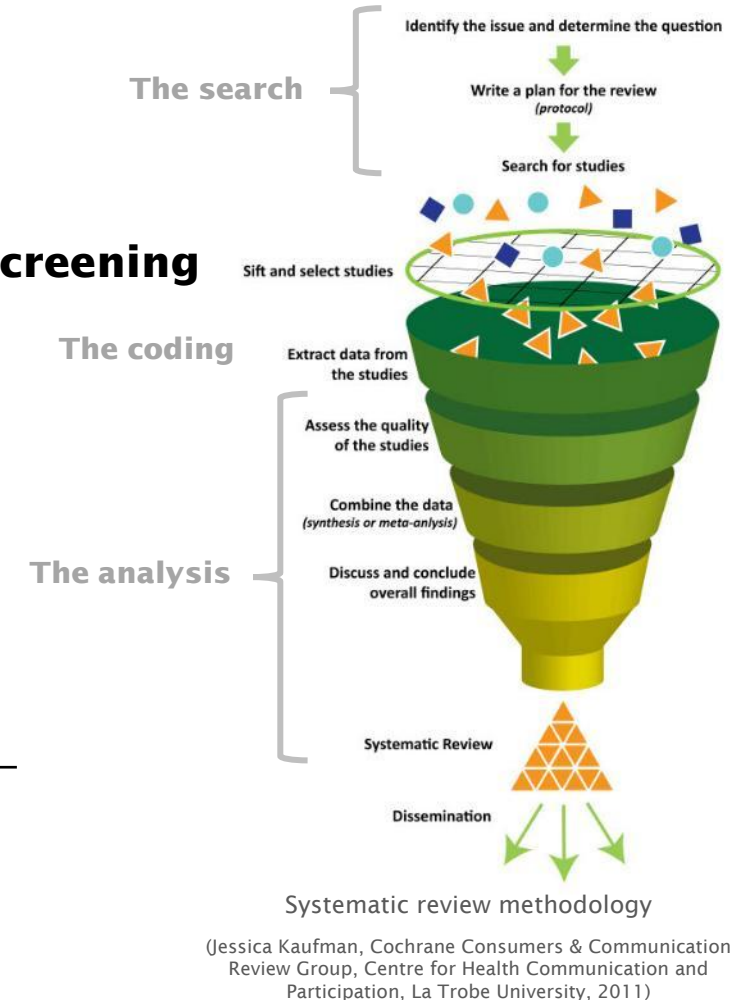
The screening phase

The software



Title_screenings	Abstract_screening	Fulltext_found	Final_decision	Duplicate	Data_extracted	Search_origin	Authors
yes	yes	yes	yes		yes	GoogleScholar	Al-Assaf, A., Albalawneh, A., Hjazin, A., Kabariti, R.,
yes	yes	yes	no			GoogleScholar	A Alassaf, D Alhunaiti, J Dick...
				yes		GoogleScholar	A Cuni-Sanchez, M Pfeifer, R Marchant, ND Burgess
no						GoogleScholar	A Troy, MA Wilson
yes	yes	yes	no			GoogleScholar	Aanderud, Z.T., Bahr, J., Robinson, D.M., Belnap, J., Campbell, T.P., Gill, R.A., McMillian, B., St. Clair, S.
no						WOS_tl_ts_423	Aanderud, ZT; Bahr, J; Robinson, DM; Belnap, J; Campbell, TP; Gill, RA; McMillian, B; Clair, SS

The screening



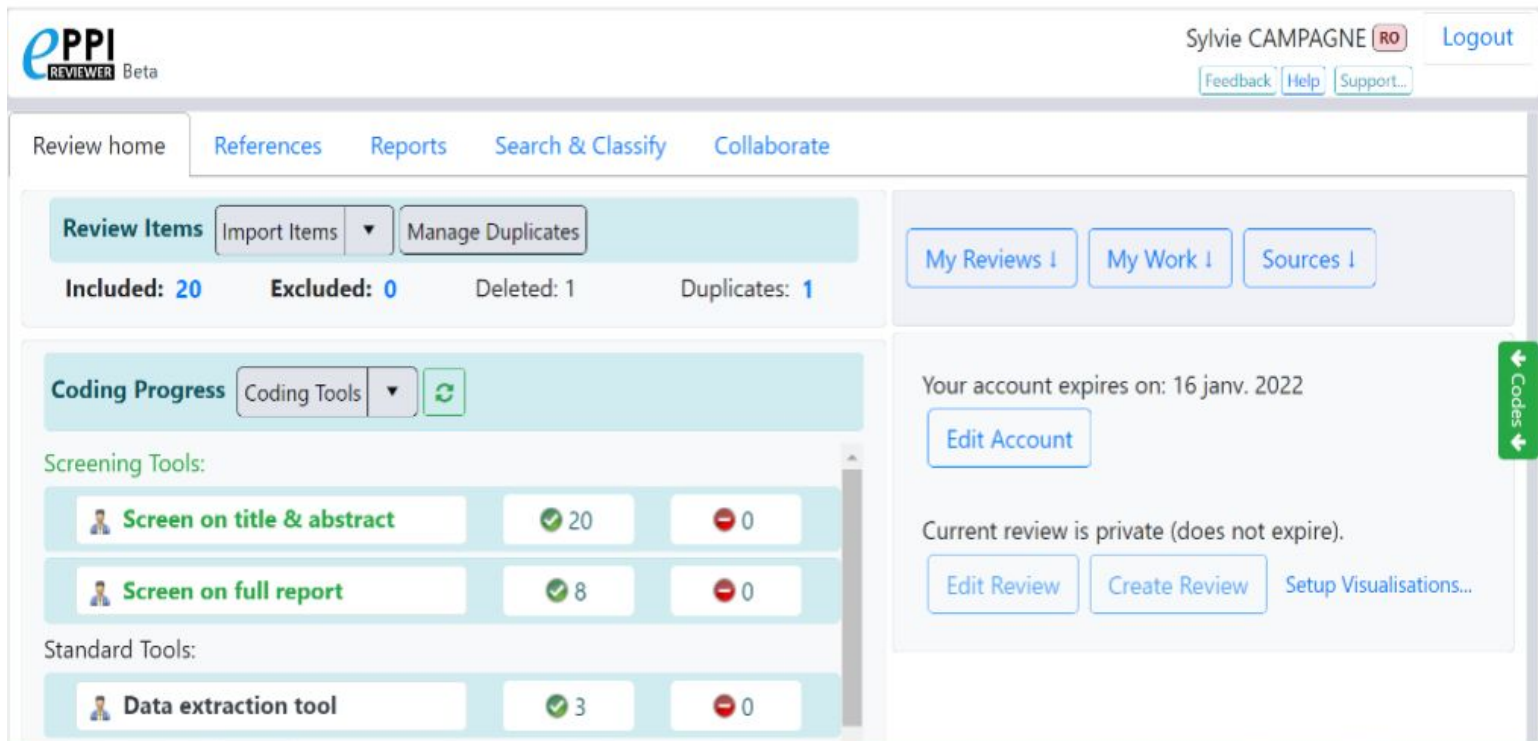
(Jessica Kaufman, Cochrane Consumers & Communication Review Group, Centre for Health Communication and Participation, La Trobe University, 2011)

The screening phase

EPPI reviewer

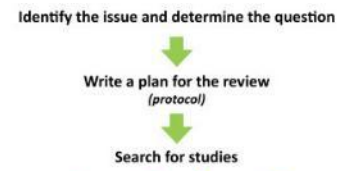
- Online tool – not free
- Very practical if several reviewers
- One place for every data

= Free version
CADIMA



The screening

The search



Sift and select studies

The coding

The analysis

Extract data from the studies

Assess the quality of the studies

Combine the data (synthesis or meta-analysis)

Discuss and conclude overall findings

Systematic Review

Dissemination

Systematic review methodology

(Jessica Kaufman, Cochrane Consumers & Communication Review Group, Centre for Health Communication and Participation, La Trobe University, 2011)

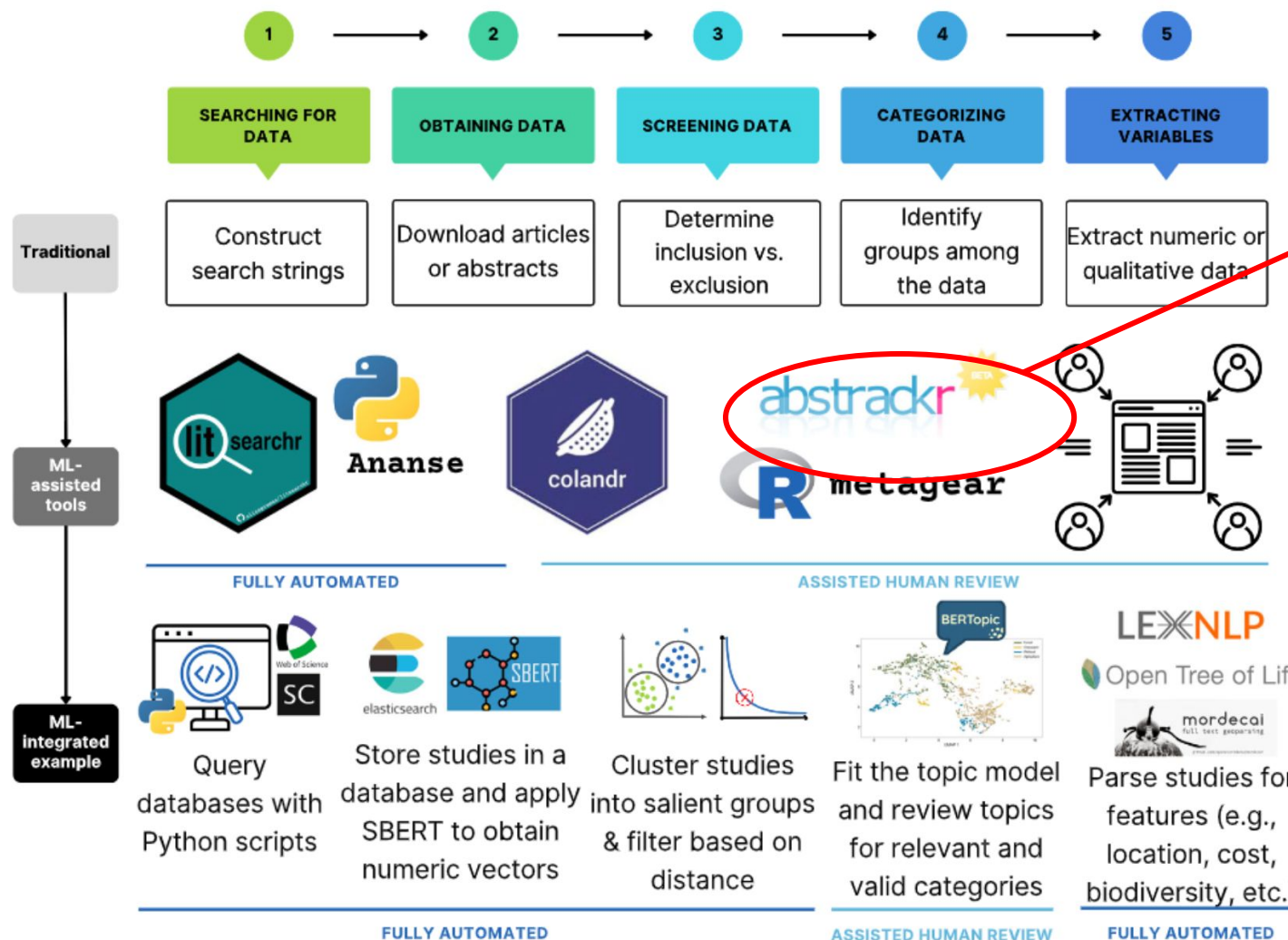
IA help



(Chang et al. 2025 New opportunities and challenges for conservation evidence synthesis from advances in natural language processing. *Conservation Biology*. 2025;39:e14464, <https://doi.org/10.1111/cobi.14464>)

The screening phase

IA help



- ✓ Free
- ✓ Easy to use
- ✓ Assist our screening phase

Semiautomated platform to screen abstracts for relevance (Gates et al., 2018; Wallace et al., 2012)

Systematic review methodology

(Chang et al. 2025 New opportunities and challenges for conservation evidence synthesis from advances in natural language processing. Conservation Biology. 2025;39:e14464, <https://doi.org/10.1111/cobi.14464>)

Agreement between different evaluators

Cohen's Kappa test for 2 raters

(see also Light's Kappa, Fleiss's Kappa)

→ Sorting results +/- disparate despite IN/OUT criteria

→ Perform assessment counts and gather them in a contingency table

Example: out of 110 articles



Damien

Jon

	YES	NO	DOUBT
YES	15	2	3
NO	0	69	8
DOUBT	0	4	9

Agreement between different evaluators

Cohen's Kappa test for 2 raters

(see also Light's Kappa, Fleiss's Kappa)

→ Calculation of Kappa

$$\kappa = \frac{P_o - P_e}{1 - P_e}$$

N: the total sum of all cells in the table

P_o: *proportion of observed agreement*, the sum of the diagonal proportions, which corresponds to the proportion of cases where the two raters assigned the same categories

P_e: *proportion of random agreement*, the sum of the products of the marginal proportions of the rows and columns

Example: Round 1 (Jon, Damien)

k = 0.68

```
# Tableau de contingence
xtab <- as.table(rbind(c(15, 2, 3), c(0, 69, 8), c(0, 4, 9)))
# Statistiques descriptives
diagonal.counts <- diag(xtab)
N <- sum(xtab)
row.marginal.props <- rowSums(xtab)/N
col.marginal.props <- colSums(xtab)/N
# Calculer kappa (k)
Po <- sum(diagonal.counts)/N
Pe <- sum(row.marginal.props*col.marginal.props)
k <- (Po - Pe)/(1 - Pe)
k
```

Agreement between different evaluators

Cohen's Kappa test for 2 raters

(see also Light's Kappa, Fleiss's Kappa)

→ Interpretation

Example: we had to discuss before a second round... :)

Less punitive: % agreement, in our case

93/110 = 85%

Value of k	Strength of the agreement
< 0	Poor
0.01 - 0.20	Light
0.21 - 0.40	Fair
0.41 - 0.60	Moderate
0.61 - 0.80	Substantial
0.81 - 1	Almost perfect

The key phases

The search

- Define the question
- Define the search terms
- Define the search string
- Define the search sources : a bibliographic database (WoS) and a web search engine (google scholar)

The screening

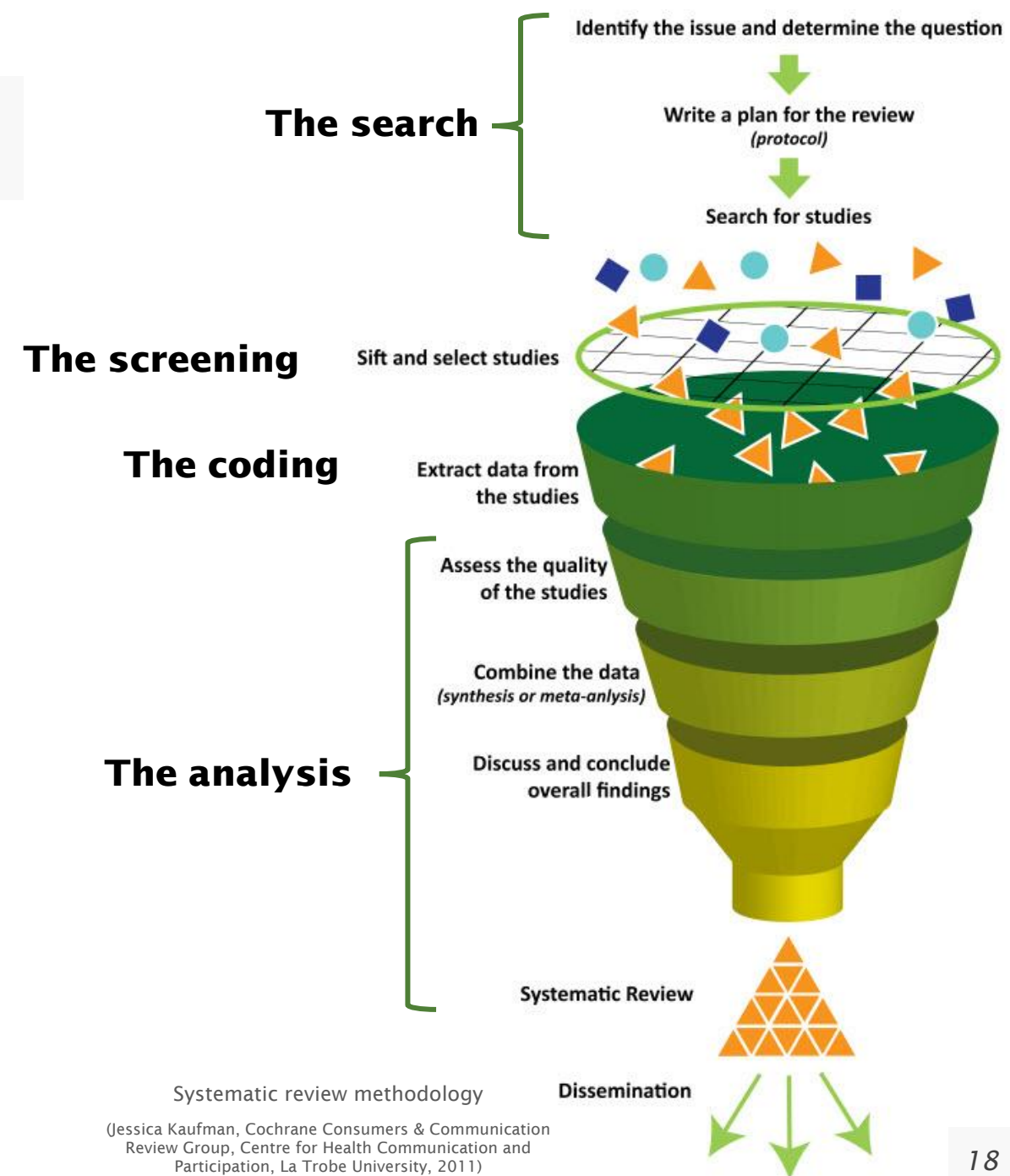
- Define the eligibility criteria

The coding

- Harmonize the data extracted

The analysis

- Show review descriptive statistics : Use standard reporting tool
- Show bibliographic information
- Show extracted data



Reporting

Meeting Name

The needs

- Transparency, rigor and traceability are key objectives of systematic maps
- In the “ classic ” reviews the details of the stages and volumes are only very rarely exposed, sometimes deductible but most often totally hidden . Decisions are not tracked .

Without reporting:

- ⇒ The review is not replicable
- ⇒ The reader cannot understand how the final result is obtained (missing of studies primary ? high rate of inaccessible pdfs , etc.)



TRACEABILITY

The needs

Light pollution: A review of the scientific literature

GeneralSearch&SID=F5ZJLB669fH5Bx5kkcR&preferencesSaved=) from the beginning of the database (2003) to 2019. A total of 8051 publications were obtained on October 3, 2020. It needs to be pointed out the authors of this review paper collectively can work in English, German, Spanish and French languages. However, we restricted our search on publications written in English only for the general scientific community, considering English as a vehicular language (Clouet, 2017; Gordin, 2015). During the literature selection, all the abstracts were assessed to select the only ones containing the information about light pollution in the introduction, methods, results or conclusions. Books were not included and only one PhD thesis, indexed in this database, was taken into account. In the end, 621 publications were used. Therefore, there may be valuable information published in other literature finders (e.g. Google Scholar), other different publication formats or distinct languages that were not considered.

The information was organised and classified in an Excel datasheet under the following criteria:

Rodrigo-Comino et al., 2021

The Anthropocene Review => AI: 3.682

- ⇒ What happened between the export and the final corpus?
- ⇒ How many articles are excluded and on what criteria?
- ⇒ How many pdfs not found?

A synthesis of two decades of research documenting the effects of noise on wildlife

II. LITERATURE REVIEW METHODOLOGY

We conducted a detailed literature search using Thompson's *ISI Web of Science* within the following subject areas 'Acoustics', 'Zoology', 'Ecology', 'Environmental Sciences', 'Ornithology', 'Biodiversity Conservation', 'Evolutionary Biology', and 'Marine Freshwater Biology' from 1990 to 2013. The specific search terms were ([WILDLIFE or ANIMAL or MAMMAL or REPTILE or AMPHIBIAN or BIRD or FISH or INVERTEBRATE] and [NOISE or SONAR]), which returned a total of 2205 scientific peer-reviewed articles. These papers were filtered so only empirical studies focussed on documenting the effects of anthropogenic noise on wildlife were included in the final data set ($N = 242$). Reviews, syntheses, method papers ($N = 32$), and studies dealing solely with natural acoustic sources ($N = 22$) were excluded.

Shannon et al., 2016

Biological Reviews => AI: 14,350

CEESAT form for overviews CEEDER assessment

4.3 Are eligibility decisions transparently reported? ^{*}

Rationale: Listing all articles that were screened for eligibility and indicating whether each was included or excluded in the synthesis (usually as supplementary material), makes it clear whether potentially relevant studies have been omitted according to the eligibility criteria or were not captured by the search. Documenting the reasons for article exclusion at text is essential for transparency.

Mark only one oval.

☐ 4 - Gold: The number of unique articles found during the searches (after removal of duplicates) is presented AND The number excluded at each stage of the screening process is fully presented (e.g. in a flow diagram or table) AND Reasons for exclusion of each article/study considered at full-text are presented (e.g. in an appendix) AND A list of eligible (included) articles/studies is presented as a separate list or in tables (not just included in reference list)

☐ 3 - Green: The number of articles excluded at each stage of the screening process is reported but some aspects missing (e.g. number of unique articles or articles unobtainable) AND Reasons for exclusion of each article/study considered at full-text are presented (e.g. in an appendix) AND A list of eligible (included) articles/studies is presented as a separate list or in tables (not just included in reference list).

☐ 2 - Amber: The number of articles excluded during the screening process is reported (or inferable) but some aspects missing (e.g. number of unique articles or articles unobtainable) AND A list of eligible (included) articles/studies is presented as a separate list or in tables (not just included in reference list)

☐ 1 - Red: No to either or both of the amber criteria above

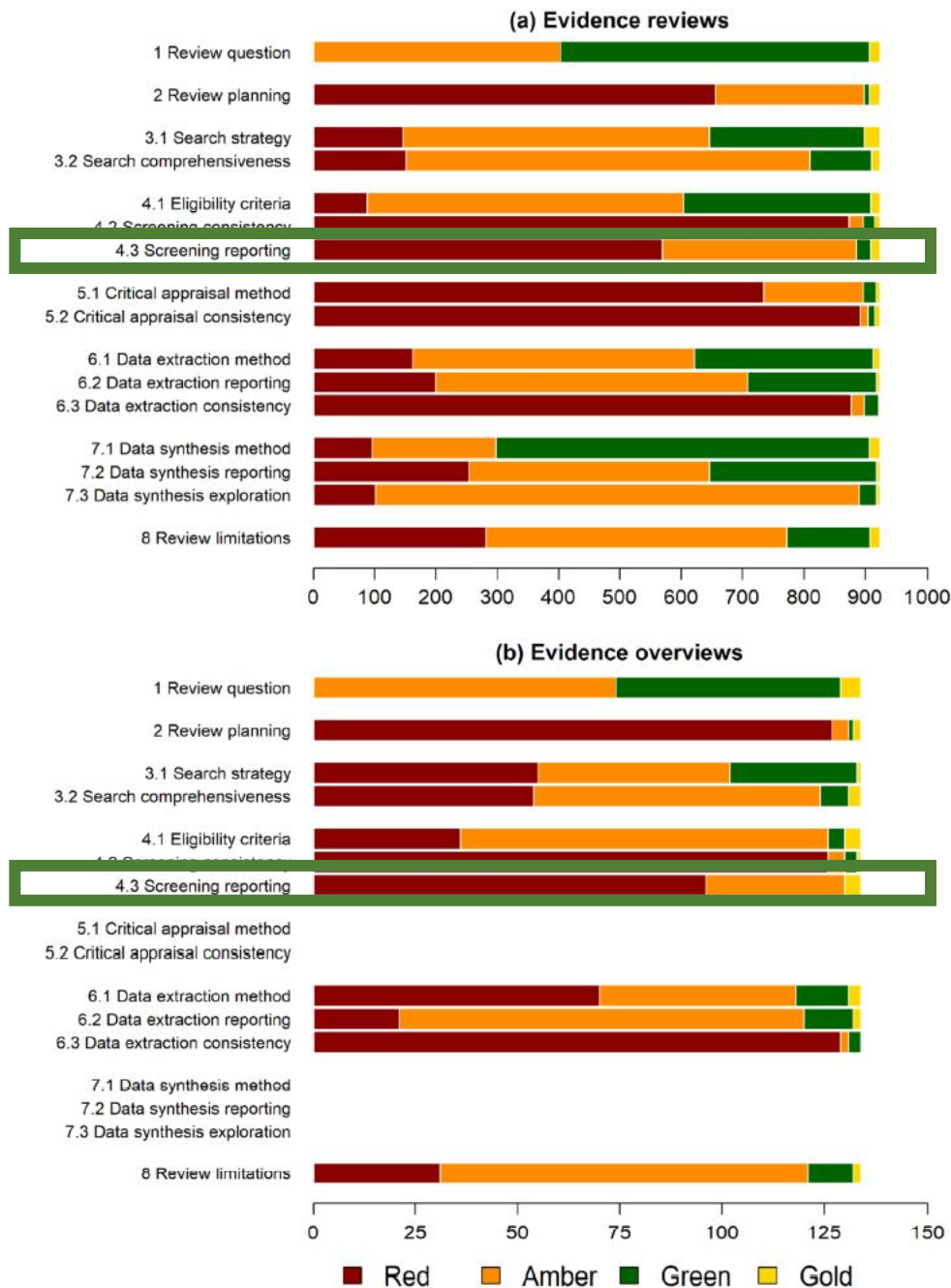


Fig. 2 The distribution of CEESAT ratings for each criterion for evidence reviews ($n=924$, top) and evidence overviews ($n=134$, bottom) published between 2018 and 2020. Note, no red category is included for Criterion 1 as this is an eligibility criterion for inclusion in the CEEDER database (red articles for criterion 1 are excluded from CEEDER). CEESAT criteria 5 and 7 are not applied to overviews



- ROSES = **RepOrting standards for Systematic Evidence Syntheses**
- Forms designed specifically for systematic reviews and maps in the field of conservation and environmental management have been produced by the EEC
- ROSES was created by a team of researchers with experience in systematic reviews in the environmental field.
- From pre-existing tools in other fields (like PRISMA in the medical field)

Methodology | [Open Access](#) | Published: 19 March 2018

ROSES RepORting standards for Systematic Evidence Syntheses: *pro forma*, flow-diagram and descriptive summary of the plan and conduct of environmental systematic reviews and systematic maps

Neal R. Haddaway, Biljana Macura , Paul Whaley & Andrew S. Pullin*Environmental Evidence* 7, Article number: 7 (2018) | [Cite this article](#)14k Accesses | 214 Citations | 34 Altmetric | [Metrics](#)

Abstract

Reliable synthesis of the various rapidly expanding bodies of evidence is vital for the process of evidence-informed decision-making in environmental policy, practice and research. With the rise of evidence-base medicine and increasing numbers of published systematic reviews, criteria for assessing the quality of reporting have been developed. First QUOROM (Lancet 354:1896–1900, 1999) and then PRISMA (Ann Intern Med 151:264, 2009) were developed as reporting guidelines and standards to ensure medical meta-analyses and systematic reviews are reported to a high level of detail. PRISMA is now widely used by a range of journals as a pre-submission checklist. However, due to its development for systematic reviews in

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Sections

[References](#)[Abstract](#)[Background](#)[PRISMA and environmental reviews](#)[Aims and objectives](#)[Methods](#)[Key differences between ROSES and PRISMA](#)[Distinction between checklist and meta-data](#)[Digitalisation of ROSES](#)[Benefits of ROSES](#)[Conclusions](#)[References](#)[Authors' contributions](#)

<https://doi.org/10.1186/s13750-018-0121-7>

Aims and scope

Fees and funding

Language editing
services

Copyright

▼ Preparing your
manuscript

Systematic
Review

Systematic Review
Protocol

Systematic Map

Systematic Map
Protocol

Methodology

Commentary

Letter to the Editor

Research Article

Evidence in Action

Prepare supporting
information

Conditions of
publication

Editorial policies

Peer review policy

Manuscript transfers

Promoting your

Systematic Review

Criteria

A systematic review is a review of evidence relevant to a clearly formulated question that uses systematic and explicit methods to identify, select and critically appraise relevant research, and to collect and analyse data from the studies that are included within the review. Authors should note that all systematic reviews published in Environmental Evidence will have been conducted according to the CEE process, including registration and publication of a protocol. Please contact the Editors at an early stage of planning your review. Full [guidelines](#) and standards can be [accessed here](#) and should have been read carefully at the protocol stage.

Note that Environmental Evidence considers it mandatory for all submitting authors to complete the relevant ROSES forms as part of their submission to demonstrate that they have included all relevant methodological details in their documents. Authors should also use the ROSES template for a flow diagram to report inclusion/exclusion process and included literature sources. Templates for ROSES forms can be accessed [here](#) and the for flow diagram [here](#). ROSES forms and flow diagram should always be downloaded from the ROSES website as it contains most up-to-date templates. Please note the formatting of your submission should follow our [guidelines](#) and not the ROSES template. The ROSES forms should be uploaded along with the submitted manuscript as a single-page supplementary file in a PDF format. Failure to do so could result in your manuscript being returned before review. In order to convert your completed ROSES form from a spreadsheet to a single-page PDF document, please scale the ROSES sheet to fit A4 landscape size.

For systematic reviews to be relevant to policy and practice they need to be as up-to-date as possible. Consequently, at the time of acceptance for publication, the search should normally be less than two years old. We therefore recommend that systematic reviews should be submitted

Official journal of



Submit manuscript

Editorial Board

Instructions for Editors

Sign up for article alerts and
news from this journal

CEE Guidelines and Standards

Article Collections

Annual Journal Metrics

Citation Impact

3.744 - 2-year Impact
factor (2021)

5.827 - 5-year Impact
factor (2021)

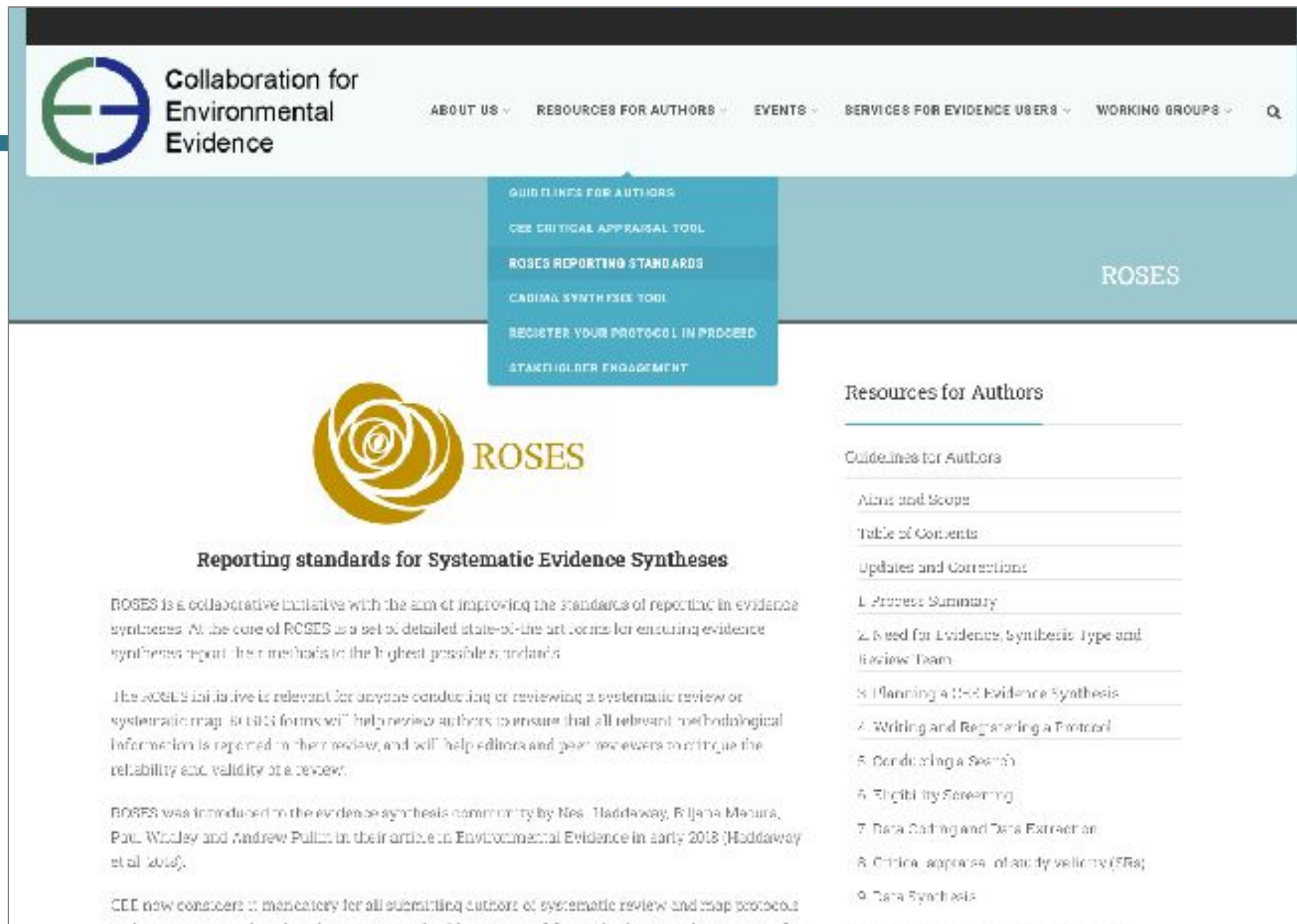
1.322 - Source Normalized
Impact per Paper (SNIP)

Environmental Evidence considers it mandatory for all authors to complete ROSES forms as part of their submission to demonstrate that they have included all relevant methodological details in their papers. Authors should also use the ROSES template for a flow chart to indicate the inclusion/exclusion process and the literature sources included.



- This is a form to fill out, with a list of details about the review/map process (number of databases, volumes, etc.)
- It is to be included as the first additional file of the manuscript (protocol & map/review), in PDF format
- The ROSES form :
 - makes manuscript writing easier for authors by allowing them to ensure they have included the right information with the right level of detail
 - ⇒ writing assistance, can avoid manuscript returns
 - ensures that all necessary content required by the CEE guidelines is present and described when submitting the manuscript
 - ⇒ EEJ control tool (“checklist”)
 - is a guarantee and information support for future readers and users of a map/magazine
 - ⇒ guarantee of transparency and rigor

!/ Always
re-download forms
to make sure you
have the latest
versions (regular
updates)



The screenshot shows the CEE website with the following elements:

- Header:** CEE logo and "Collaboration for Environmental Evidence" text. Navigation links: ABOUT US, RESOURCES FOR AUTHORS, EVENTS, SERVICES FOR EVIDENCE USERS, WORKING GROUPS.
- Left Sidebar:**
 - GUIDELINES FOR AUTHORS
 - CEE CRITICAL APPRAISAL TOOL
 - ROSES REPORTING STANDARDS (highlighted)
 - CADIMA SYNTHESIS TOOL
 - REGISTER YOUR PROTOCOL IN PROCEED
 - STAKEHOLDER ENGAGEMENT
- Main Content Area:**
 - ROSES logo (a stylized rose).
 - Reporting standards for Systematic Evidence Syntheses**
 - Text: "ROSES is a collaborative initiative with the aim of improving the standards of reporting in evidence syntheses. At the core of ROSES is a set of detailed state-of-the-art forms for ensuring evidence syntheses report their methods to the highest possible standards."
 - Text: "The ROSES initiative is relevant for anyone conducting or reviewing a systematic review or systematic map. ROSES forms will help review authors to ensure that all relevant methodological information is reported in their review, and will help editors and peer reviewers to critique the reliability and validity of a review."
 - Text: "ROSES was introduced to the evidence synthesis community by Neil Haddaway, Biljana Manura, Paul Whitley and Andrew Pullin in their article in Environmental Evidence in early 2018 (Haddaway et al 2018)."
 - Text: "CEE now considers it mandatory for all submitting authors of systematic review and map protocols and reports to complete the ROSES forms as part of their submission to ensure that the highest standards are maintained."
- Right Sidebar:**
 - Resources for Authors**
 - Guidelines for Authors
 - Aims and Scope
 - Table of Contents
 - Updates and Corrections
 - 1. Process Summary
 - 2. Need for Evidence, Synthesis Type and Review Team
 - 3. Planning a New Evidence Synthesis
 - 4. Writing and Registering a Protocol
 - 5. Conducting a Search
 - 6. Eligibility Screening
 - 7. Data Coding and Data Extraction
 - 8. Critical Appraisal of study validity (SRA)
 - 9. Data Synthesis
 - 10. Dissemination and Reporting

!/ Toujours aller
retélécharger les
formulaire pour
être sûrs d'avoir les
dernières versions
(MAJ régulières)



Collaboration for Environmental Evidence

ABOUT US ▾

RESOURCES FOR AUTHORS ▾

EVENTS ▾

SERVICES FOR EVIDENCE USERS ▾

WORKING GROUPS ▾



not require a considerable time commitment, but if done early, could save considerable resources by avoiding systematic reports being rejected by editorial staff prior to peer review.

CHE encourages authors to read ROSES checklists for reviews and reports when so they begin a new project to facilitate completion of the forms at the submission stage. The information and support in the ROSES forms could also make the job of conducting a review and drafting the protocol and report documents much easier.

Authors are encouraged to check the ROSES website to ensure they are using the most up-to-date version of the ROSES forms.

Below are direct links to information about ROSES and ROSES forms for all CHE review authors.

ROSES for systematic map protocols

ROSES for systematic map reports

ROSES for systematic review protocols

ROSES for systematic review reports

ROSES template for flow diagram of review activities

ROSES website

Haddaway et al. 2018 ROSES Reporting standards for Systematic Evidence Syntheses: pro forma, flow-diagram and descriptive summary of the plan and conduct of environmental systematic reviews and systematic maps. Environmental Evidence

EE Critical Appraisal Tool

ROSES reporting standards

CADMA Synthesis Tool

PROOF

Stakeholder Engagement



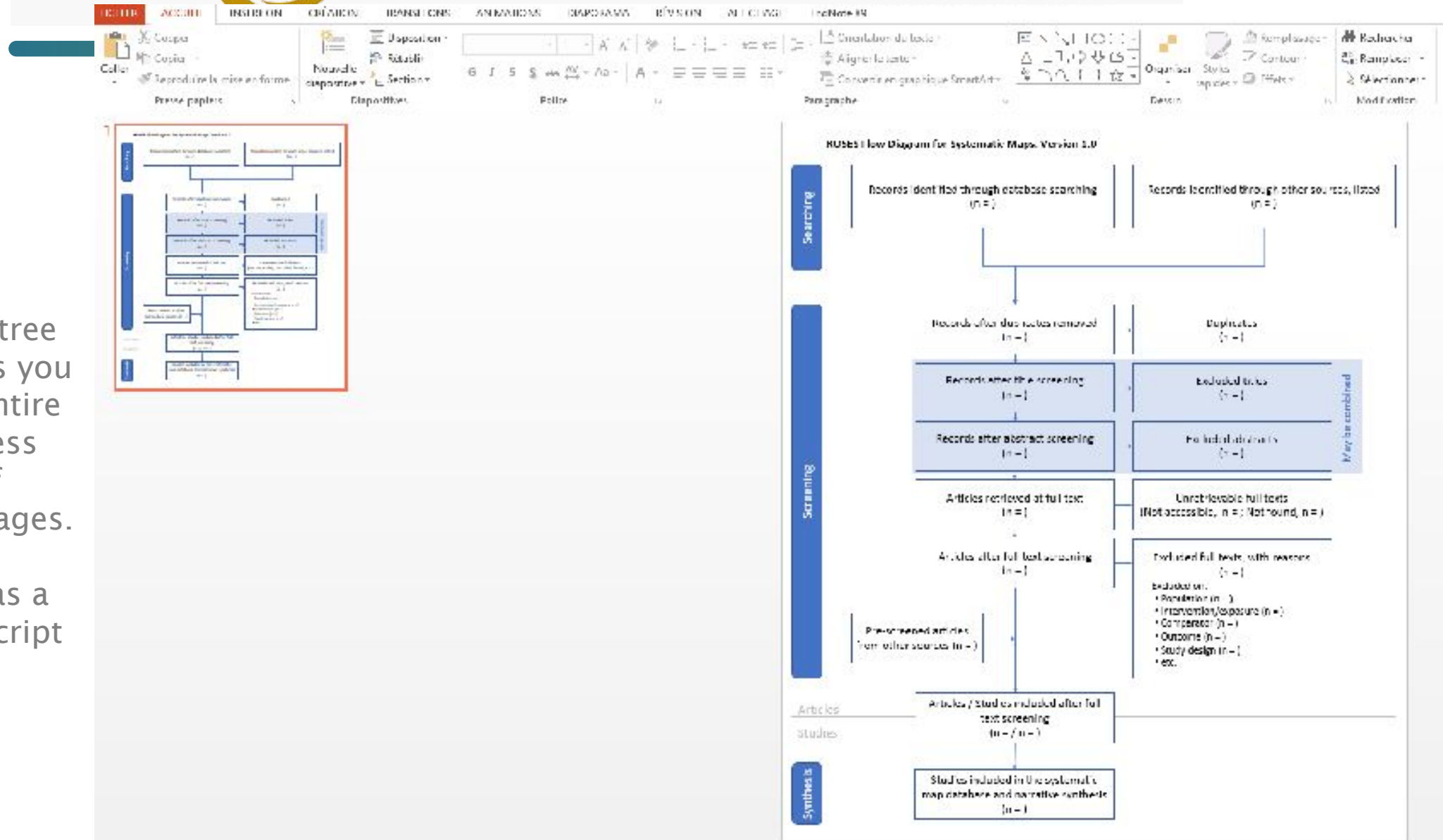
<https://environmentalevidence.org/roses/>



ROSES Flow diagram for systematic maps

A flow diagram is a tree structure that allows you to quickly see the entire map or review process and the evolution of volumes over the stages.

It is to be included as a figure in the manuscript of a map/review





ROSES
Online tool for
producing ROSES
flow diagrams for
systematic maps
and reviews
https://estech.shinyapps.io/roses_flowchart/

Options

Synthesis type:

review

Include prescreened records:

yes

Title and abstract screening:

combined

Select type of synthesis:

narrative

Data sources

Database results:

Other sources results:

Prescreened records:

Additional description of prescreened studies:

Deduplication

Deduplicated records:

Duplicates removed:

Screening

Included titles and abstracts:

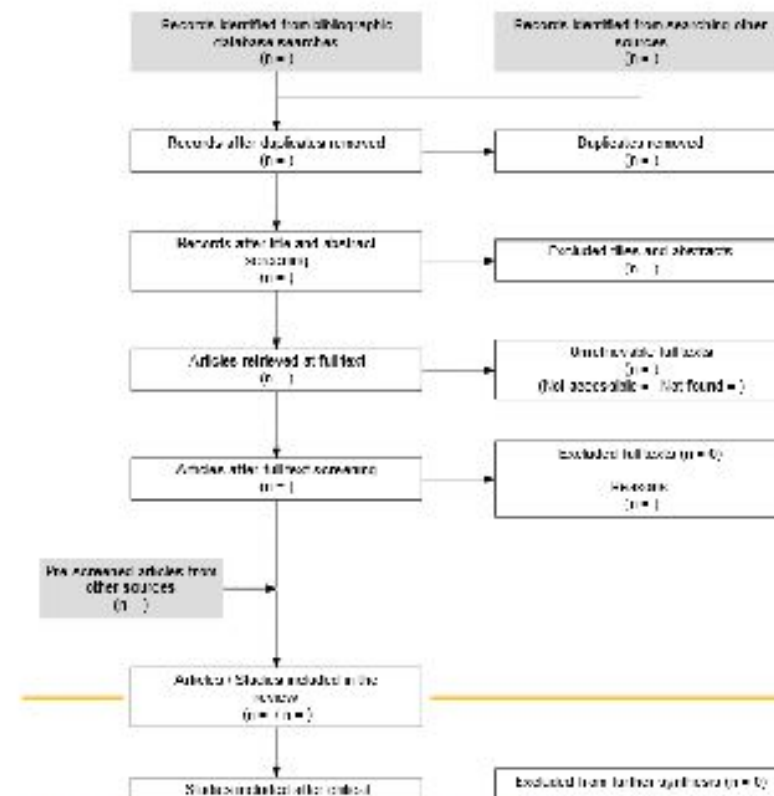
Excluded titles and abstracts:

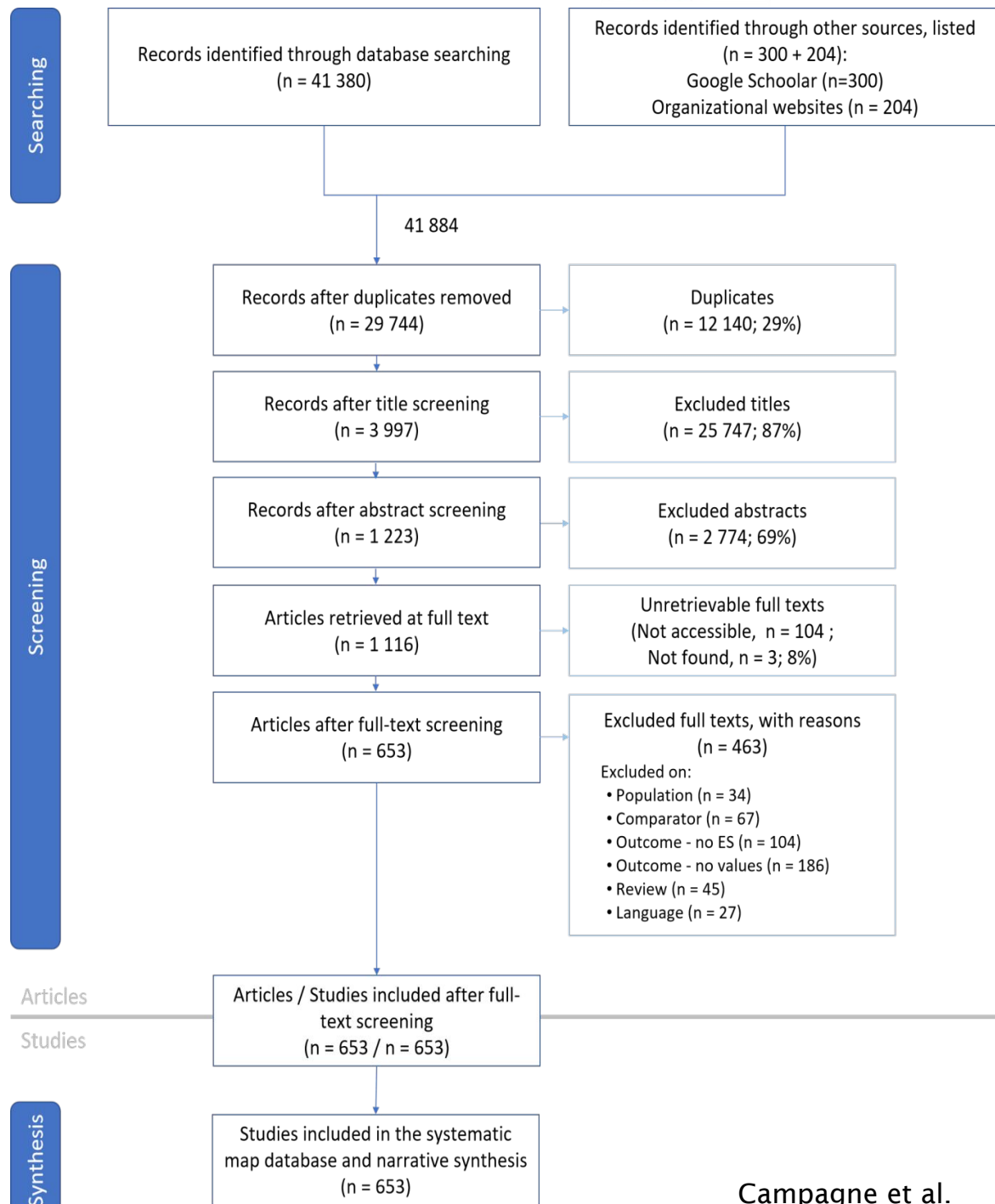
Retrieved full texts:



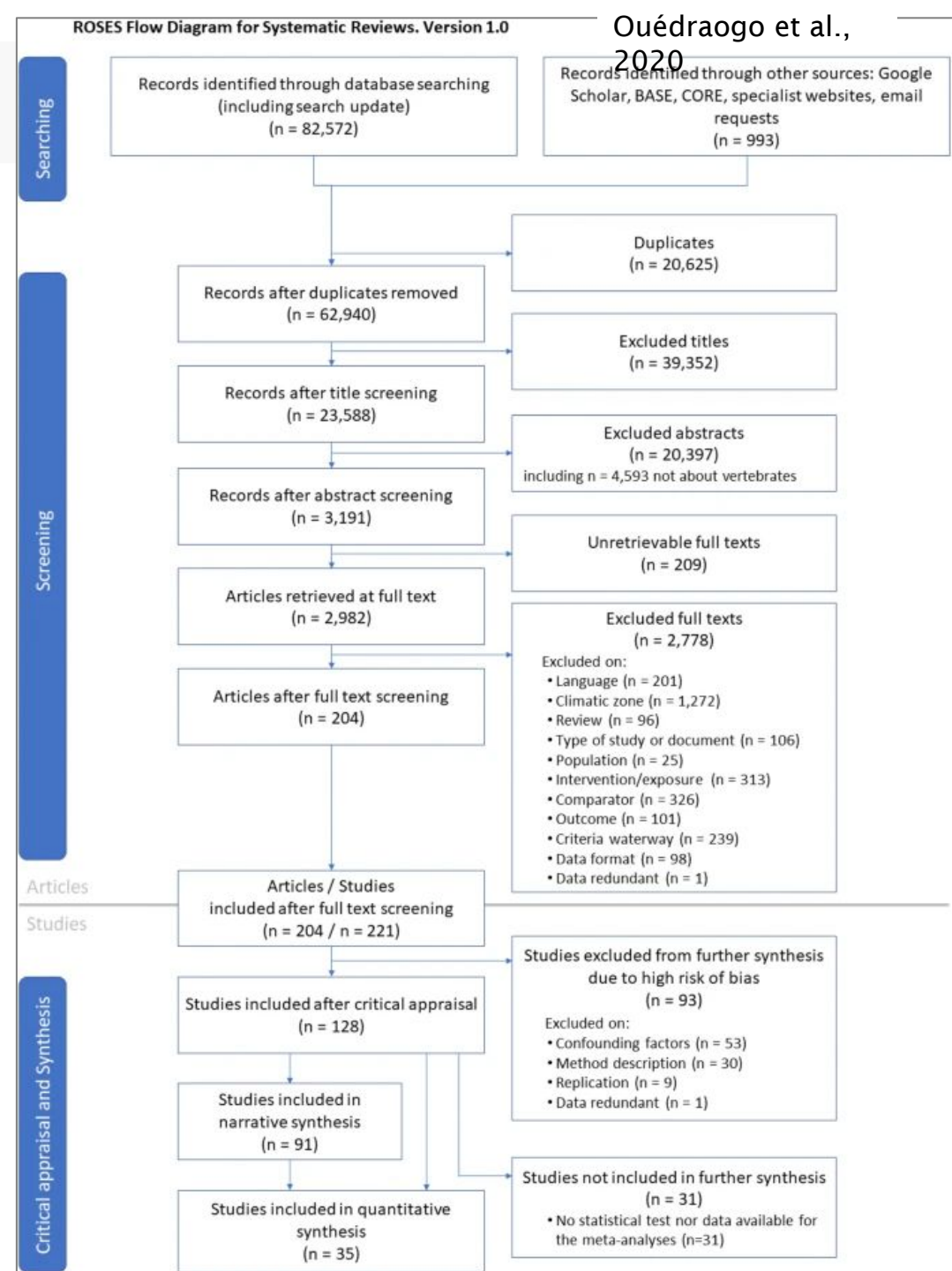
ROSES

Reporting standards for Systematic Evidence Syntheses





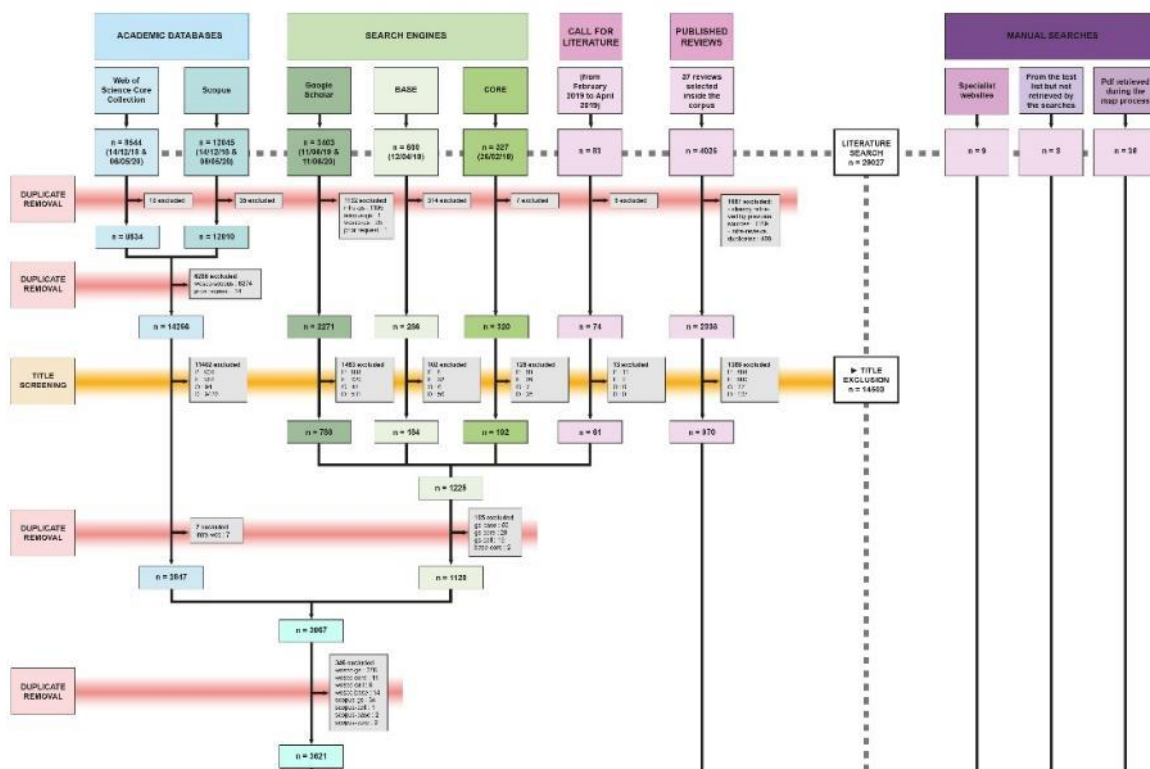
Campagne et al.
2023



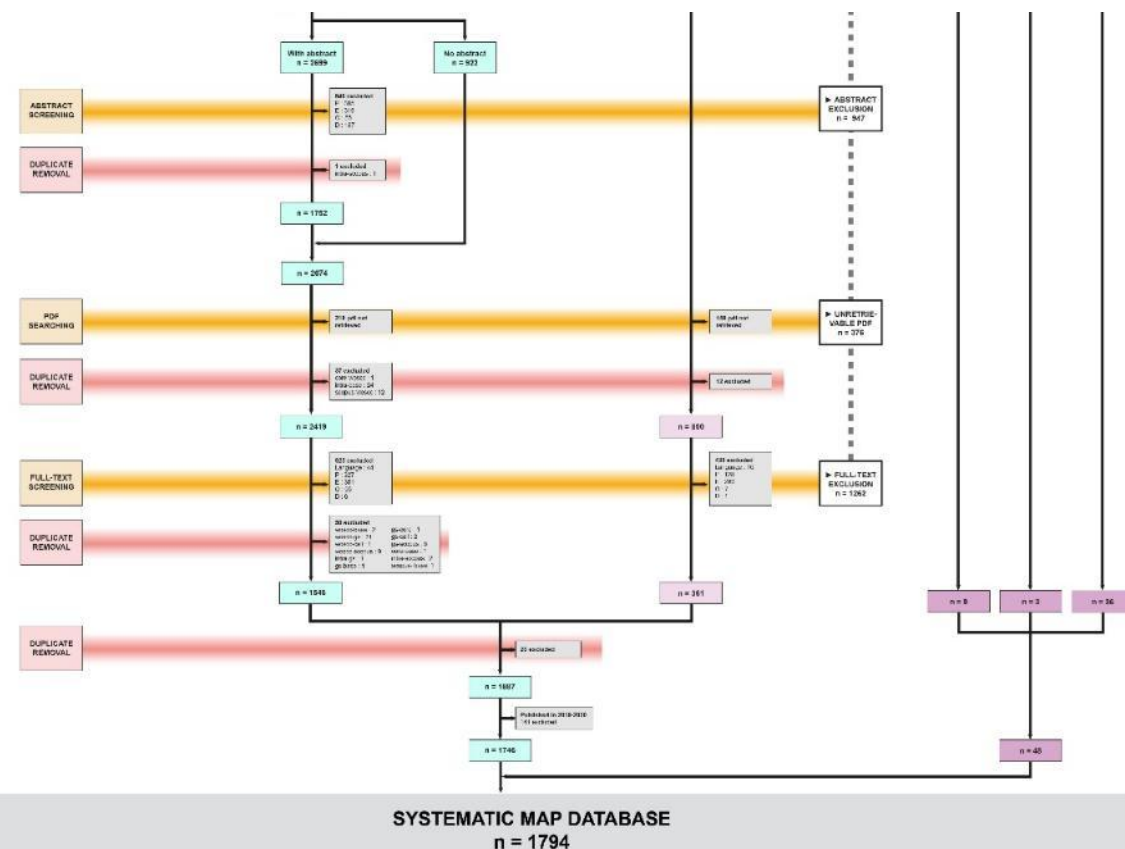
Ouédraogo et al.,
2020

More detailed flow diagram

SEARCHING



SCREENING



Sordello et al., 2020 (additional file)

ROSES: the form for systematic reviews

ROSES: the form for systematic reviews

Formin: SORDELLO

Item number	Section/sub-section	Topic	Description	Further explanation	Checklist/meta-data	Author response	Comments
1	Title	Title	The title must indicate that it is a systematic review, and should indicate	The title should normally be the same or very similar to the review	Meta-data		
2	Type of review	Type of review	Select one of the following types of review: systematic review.	See CEE Guidance on amendments and updates [1]	Meta-data		
3	Authors' contacts	Authors' contacts	The full names, institutional addresses and email addresses for all		Checklist		
4	Abstract	Structured summary	The abstract of the manuscript must not exceed 500 words and must be		Checklist		
5	Background	Background	Describe the rationale for the review in the context of what is already	A theory of change and/or conceptual model should be presented that	Checklist		
6	Stakeholder engagement	Stakeholder engagement	The actual role of stakeholders throughout the review process (e.g. in		Checklist		
7	Objective of the review	Objective	Describe the primary question and secondary questions (when	The primary question is the main question of the review. The secondary	Checklist		
8	Definition of the question	Definition of the question	Provide reference to the question key elements, e.g. population(s).	For other question types see [3,4]	Meta-data		
9	Methods	Protocol	Provide citation, DOI or open access link to published protocol.	The protocol should be peer reviewed and publicly available online	Meta-data		
10	Deviations from protocol	Deviations from protocol	Describe any ways in which the final methods of the review deviate from		Checklist		
11	Searches	Search strategy	Detail the search strategy used, including database names searched.		Checklist		
12	Search string	Search string	Provide Roam-style full search string and state the platform for which		Meta-data		
13	Languages – bibliographic	Languages – bibliographic	List languages used in bibliographic database searches		Meta-data		
14	Languages – grey literature	Languages – grey literature	List languages used in organisational website searches and web based		Meta-data		
15	Bibliographic databases	Bibliographic databases	Provide the number of bibliographic databases searched		Meta-data		
16	Web-based search engines	Web-based search engines	Provide the number of web-based search engines searched		Meta-data		
17	Organisational websites	Organisational websites	Provide the number of organisational websites searched		Meta-data		
18	Estimating comprehensiveness of	Estimating comprehensiveness of	Describe the process by which the comprehensiveness of the search	Compulsory (if update performed). A search update is good practice if	Checklist		
19	Search update	Search update	Describe any update to searches undertaken during the conduct of the		Checklist		
20	Article screening and study	Screening strategy	Describe the methodology for screening articles/studies for relevance.		Checklist		
21	Inclusion criteria	Inclusion criteria	Describe the inclusion criteria used to assess relevance of identified		Checklist		
22	Critical appraisal	Critical appraisal strategy	Describe here the method used for critical appraisal of study validity		Checklist		
23	Critical appraisal used in synthesis	Critical appraisal used in synthesis	Describe how the information from critical appraisal was used in		Checklist		
24	Data extraction	Meta-data extraction and coding	Describe the method for meta-data extraction and coding for studies.	Optional, a map database can be included within a systematic review	Checklist		
25	Data extraction strategy	Data extraction strategy	Describe the method for extraction of qualitative and/or quantitative		Checklist		
26	Approaches to missing data	Approaches to missing data	Describe any process for obtaining and confirming missing or unclear		Checklist		
27	Potential effect modifiers/reasons	Potential effect modifiers/reasons	Provide a list of and justification for the effect modifiers/reasons for		Checklist		
28	Data synthesis and presentation	Type of synthesis	State the type of synthesis conducted as part of the systematic review		Meta-data		
29	Narrative synthesis strategy	Narrative synthesis strategy	Describe methods used for narratively synthesising the evidence base in		Checklist		
30	Quantitative synthesis strategy	Quantitative synthesis strategy	If data are appropriate for quantitative synthesis, describe methods for	Compulsory (if quantitative synthesis performed)	Checklist		
31	Qualitative synthesis strategy	Qualitative synthesis strategy	Describe methods used for synthesising qualitative data and justify your	Compulsory (if qualitative synthesis performed)	Checklist		
32	Other synthesis strategies	Other synthesis strategies	Describe any other approaches used for synthesising data or combining	Compulsory (if other synthesis performed)	Checklist		
33	Assessment of risk of publication	Assessment of risk of publication	Describe methods for examining the possible influence of publication	This may be done for quantitative syntheses using diagnostic plots on	Checklist		
34	Knowledge gap and cluster	Knowledge gap and cluster	Describe the methods used to identify and/or prioritise key knowledge	Optional	Checklist		
35	Diminishing procedural	Diminishing procedural	Describe the role of systematic reviewers (who have also authored	Reviewers who have authored articles to be considered within the	Checklist		
36	Description of review process	Description of review process	Describe the review process including the volume of evidence identified		Checklist		

ROSES: the form for systematic reviews



ROSES for Systematic Review Reproducible - Form 1

Summary: SORDELLO

	C	D	E	F	G	H	
1 Topic		Description		Further explanation	Checklist/meta-data	Author response	Comments
2 Title		The title must indicate that it is a systematic review, and should indicate if		The title should normally be the same or very similar to the review	Meta-data		
3 Type of review		Select one of the following types of reviews: systematic review, systematic		See CEB Guidance on amendments and updates [1]	Meta data	systematic review	
4 Authors' contacts		The full names, institutional addresses and email addresses for all authors			Checklist	Yes	
5 Structured summary		The abstract of the manuscript must not exceed 500 words and must be			Checklist	Yes	
6 Background		Describe the rationale for the review in the context of what is already		A theory of change and/or conceptual model should be presented that	Checklist	Yes	
7 Stakeholder engagement		The actual role of stakeholders throughout the review process (e.g. in			Checklist	Yes	
8 Objective		Describe the primary question and secondary questions (when applicable).		The primary question is the main question of the review. The secondary	Checklist		
9 Definition of the question		Provide reference to the question key elements, e.g. population(s).		For other question types see [3,4]	Meta data		
10 Protocol		Provide citation, DOI or open-access link to published protocol.		The protocol should be peer-reviewed and publicly available online (open	Meta-data		

Please enter only your own

- **PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analyses**
- PRISMA is an evidence-based minimum set of items for reporting in systematic reviews and meta-analyses
- Développement des reporting dès les années 1990 : Moher D, Cook DJ, Eastwood S, Olkin I, Rennie D, et al. for the QUOROM group (1999) Improving the quality of reporting of meta-analysis of randomized controlled trials: The QUOROM statement. Lancet 354: 1896-1900.
- En 2009, mise à jour des guidelines pour tenir compte de plusieurs avancées conceptuelles et pratiques dans le domaine des revues systématiques et a été rebaptisée PRISMA (Preferred Reporting Items of Systematic reviews and Meta-Analyses).

=> PRISMA Flow diagram (équivalent du ROSES Flow diagram)

=> PRISMA checklist (équivalent du ROSES Form)



Welcome to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) website!

PRISMA is an evidence-based minimum set of items for reporting in systematic reviews and meta-analyses. PRISMA primarily focuses on the reporting of reviews evaluating the effects of interventions, but can also be used as a basis for reporting systematic reviews with objectives other than evaluating interventions (e.g. evaluating aetiology, prevalence, diagnoses or prognosis).

Who should use PRISMA?

- **Authors:** PRISMA aims to help authors improve the reporting of systematic reviews and meta-analyses.
- **Journal Peer reviewers and editors:** PRISMA may also be useful for critical appraisal of published systematic reviews, although it is not a quality assessment instrument to gauge the quality of a systematic review.

News Feed

PRISMA Website re-design

The PRISMA website underwent a much-needed update in October 2019 to update the content of the website. We have updated the look of the site and added the PRISMA extensions, translations, and information about review protocols.

PRISMA Extensions!

Several PRISMA extensions have been published in 2019 so far:

- **PRISMA-P** (for developing review protocols) was published in January 2019 in *Systematic Reviews* and the *Lancet*.
- **PRISMA-PD** (individual patient data) was published in *JAMA* in April.
- **PRISMA-MMA** (Network Meta-Analysis) was published in *Annals of Internal Medicine* in June.

These are in addition to the PRISMA Abstract and Equity extensions, all found on the PRISMA website, [here](#).

[Read more](#)

Key Documents

- [PRISMA 2020 Checklist](#)
- [PRISMA 2020 flow diagram](#)
- [PRISMA 2020 Statement](#)
- [PRISMA 2020 Explanation and Elaboration](#)



PROSPERO

International prospective register of systematic reviews



equator
network

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Chris Pritchard @chriscritchard · Jul 3

If you want to check out the newest features, head on over to [extendingyourprismaflowdiagram.com](#), we now support reporting of individual databases and registers, meaning you can produce [PRISMA 5](#) compliant flow diagrams! [@PRISMASearch](#) [@nealhardaway](#) [@mcguinla](#) [@mjpages](#)

<https://prisma-statement.org/>



PRISMA
TRANSPARENT REPORTING OF SYSTEMATIC REVIEWS AND META-ANALYSES

HOME PRISMA STATEMENT EXTENSIONS TRANSLATIONS PROTOCOLS ENDORSEMENT News

PRISMA Statement PRISMA 101 Checklist **Flow Diagram** History & Development Funding Citing & Using PRISMA

PRISMA Flow Diagram

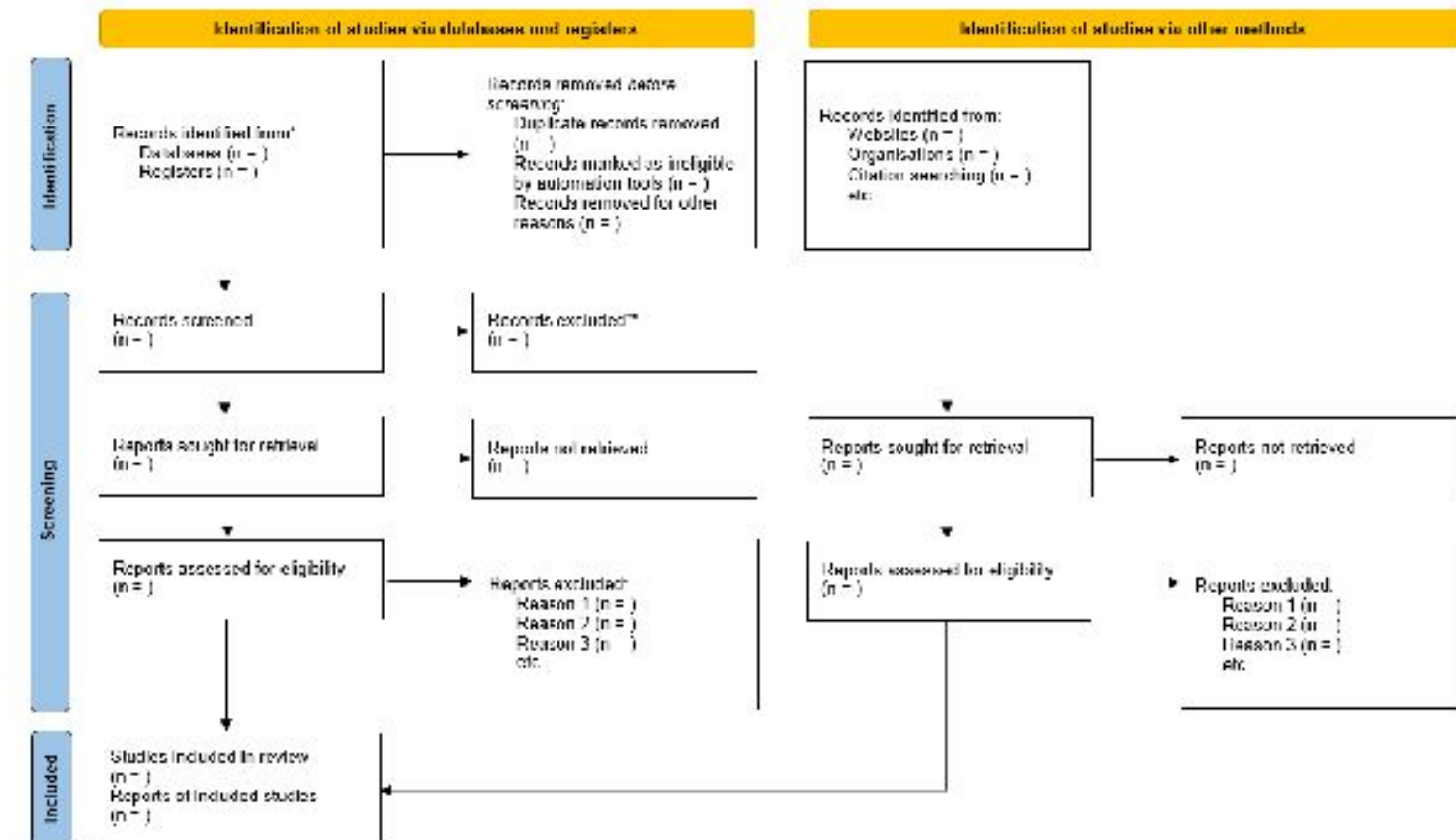
The flow diagram depicts the flow of information through the different phases of a systematic review. It maps out the number of records identified, included and excluded, and the reasons for exclusions. Different templates are available depending on the type of review (new or updated) and sources used to identify studies.

-  PRISMA 2020 flow diagram for new systematic reviews which included searches of databases and registers only
-  PRISMA 2020 flow diagram for new systematic reviews which included searches of databases, registers and other sources
-  PRISMA 2020 flow diagram for updated systematic reviews which included searches of databases and registers only
-  PRISMA 2020 flow diagram for updated systematic reviews which included searches of databases, registers and other sources

Flow diagrams can also be generated using a Shiny App available at <https://www.cshar.kathon.org/software/PRISMA2020.html>

For more information about citing and using PRISMA click here

<https://www.prisma-statement.org/PRISMAStatement/FlowDiagram>



*If automation tools were used, indicate how many records were excluded by a human and how many were excluded by automation tools.

From: Page MJ, McKenzie JL, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow GD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;372:n71. doi: 10.1136/bmj.n71. For more information, visit: <http://www.prisma-statement.org/>

Systematic reviews should be described in a high degree of methodological detail. [The PRISMA Statement](#) calls for a high level of reporting detail in systematic reviews and meta-analyses. An integral part of the methodological description of a review is a flow diagram.

This tool allows you to produce a flow diagram for your own review that conforms to [the PRISMA2020 Statement](#). You can provide the numbers in the data entry section of the 'Create flow diagram' tab. Alternatively, to allow for more customisation, you can use the template file below.

This tool also allows you to download an interactive HTML version of the plot, alongside several other common formats.

We also provide an R package: [PRISMA2020 flow diagram R package](#) on [Github](#).

Please let us know if you have any feedback or if you encounter an error by creating an issue on [Github](#).

[Download the template CSV file](#)

Upload your edited file here:

Choose CSV File

Browse

No file selected

Please cite as:

Haddaway, N. R., Page, M. J., Pritchard, C. C., & McGuinness, L. A. (2022). PRISMA2020: An R package and Shiny app for producing PRISMA 2020-compliant flow diagrams, with interactivity for optimised digital transparency and Open Synthesis. *Gampbell Systematic Reviews*, 18, e1290. <https://doi.org/10.1002/GM.1290>

[Download citation \(.ris\)](#)

Credits

Neal R. Haddaway (creator, author)

Luke A. McGuinness (coder, author)

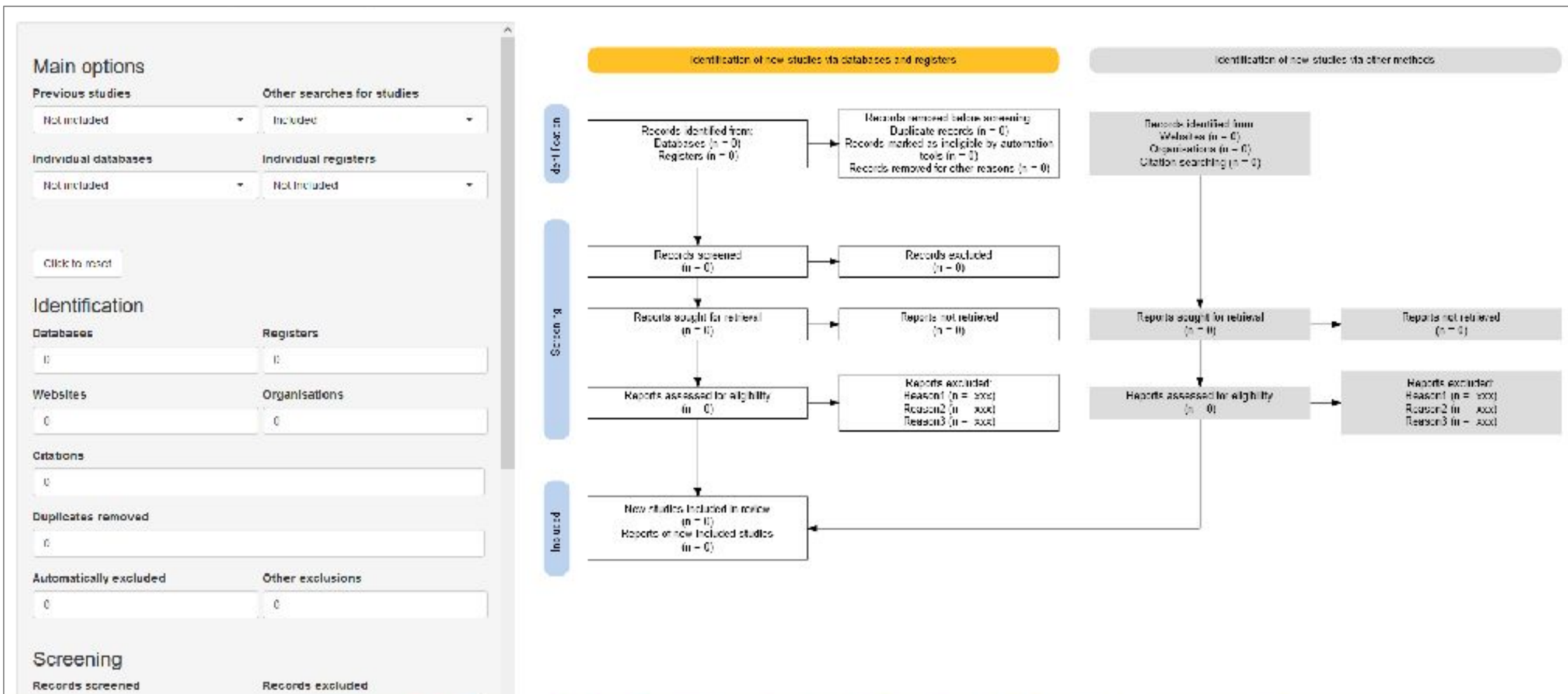
Chris C. Pritchard (coder, author)

Matthew J. Page (advisor)

Jack Wacey (advisor)



Created November 2020, Updated June 2022



https://estech.shinyapps.io/prisma_flowdiagram/



PRISMA

TRANSPARENT REPORTING OF SYSTEMATIC REVIEWS AND META-ANALYSES

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PRISMA Checklist

The PRISMA 2020 statement comprises a 27-item checklist addressing the introduction, methods, results and discussion sections of a systematic review report.



PRISMA 2020 Checklist (PDF)



PRISMA 2020 Checklist (Word)

The checklist can also be completed using a Shiny App available at <https://prisma.shinyapps.io/checklist/>

An expanded checklist, which comprises an abridged version of the reporting recommendations presented in the Explanation and Elaboration paper, with references and some examples removed, is also available.



PRISMA 2020 Expanded Checklist (PDF)

For more information about citing and using PRISMA click [here](#).

<https://www.prisma-statement.org/PRISMAStatement/Checklist>



PRISMA 2020 Checklist

Section and Topic	Item #	Checklist Item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources assessed or consulted to identify studies. Specify the date when each source was last searched or consulted.	
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (Item #4)).	
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	
	13d	Describe any methods used to synthesise results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesised results.	
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (missing from reporting biases).	
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	

Traceability of decisions : details of exclusion

**At the very least: the list of excluded full-texts with the reason.
If possible, include all items and decisions at all sorting stages.**

A	B	C	D	E	F	G	H
1	Additional File 7						
2							
3	Inclusion/exclusion decisions at the three screening stages and extraction of rejected full-texts						
4							
5	Sordello et al. 2020						
6							
7	How to read the different columns is explained in the CodeBook sheet						
8							
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	A	B	C	D	E	F	G	H
1	iblio_interna	biblio_pid	biblio_authors	biblio_cor	biblio_title	biblio_ye	exclusion_justif	justif_peco_details
2	31	10.1002/aqc.1015	Currey, RJC; Dawson	AQUATIC CO	Survival rates for a declining population of bottlenose dolphins in Dou	2009	PECO	E
3	39	10.1002/aqc.2416	Pleslic, G; Gospic, N	AQUATIC CO	The abundance of common bottlenose dolphins (<i>Tursiops truncatus</i>) in	2015	PECO	E
4	44	10.1002/aqc.2833	Braulik, GT; Kasuga	AQUATIC CO	Cetacean rapid assessment: An approach to fill knowledge gaps and ta	2018	PECO	E
5	77	10.1002/ece3.1847	Koper, N; Leston, L	ECOLOGY AN	Effects of ambient noise on detectability and localization of avian song	2016	PECO	P
6	78	10.1002/ece3.2335	Vaugoyeau, M; Adria	ECOLOGY AN	Interspecific variation in the relationship between clutch size, laying d	2016	PECO	E
7	84	10.1002/ece3.2699	Donovan, CR; Harris	ECOLOGY AN	A simulation approach to assessing environmental risk of sound expos	2017	PECO	E
8	94	10.1002/ecs2.1905	Wang, JW; Poh, CH	ECOSPHERE	Building biodiversity: drivers of bird and butterfly diversity on tropical u	2017	PECO	E
9	115	10.1002/JEMT.10185	Gesi, M; Lenzi, P; Fo	MICROSCOPY	Effects of loud noise exposure on mouse myocardium: A comparison wit	2002	PECO	P
10	135	10.1002/jwmg.21179	Green, AW; Aldridge	JOURNAL OF	Investigating impacts of oil and gas development on greater sage-grou	2017	PECO	E
11	159	10.1002/ps.4619	Mazzoni, V; Gordon	PEST MANAG	Design of a candidate vibrational signal for mating disruption against t	2017	PECO	E
12	173	10.1002/we.2160	Hu, CC; Albertani, R	WIND ENER	Wind turbine sensor array for monitoring avian and bat collisions	2018	PECO	E
13	178	10.1002/wsb.546	Rosa, P; Swider, CR	WILDLIFE SO	Disentangling Effects of Noise from Presence of Anthropogenic Infrastru	2015	PECO	E
14	197	10.1006/anbe.1999.1278	Nelson, BS	ANIMAL BEH	Avian dependence on sound pressure level as an auditory distance cue	2000	PECO	E
15	203	10.1006/appe.1996.0015	Krebs, H; Macht, M	APPETITE	Effects of stressful noise on eating and non-eating behavior in rats	1996	PECO	P
16	247	10.1007/978-1-4419-7311-1	Patricio, S	EFFECTS OF	Underwater Noise Effects From Wave Energy Devices on Marine Mamma	2012	PECO	P
17	248	10.1007/978-1-4419-7311-1	Wright, AJ	EFFECTS OF	Noise-Related Stress and Cumulative Impact Assessment	2012	PECO	O
18	250	10.1007/978-1-4419-7311-1	Breitzke, M; Bohlen	EFFECTS OF	Modeling Cumulative Sound Exposure Along a Seismic Line to Assess th	2012	PECO	P
19	257	10.1007/978-1-4419-7311-1	Mountain, DC; Ande	EFFECTS OF	The ESME Workbench: Simulating the Impact of Anthropogenic Sound or	2012	PECO	P
20	266	10.1007/978-1-4419-7311-1	Mueller-Blenkle, C	EFFECTS OF	A Novel Field Study Setup to Investigate the Behavior of Fish Related to	2012	PECO	O
21	272	10.1007/978-1-4939-2981-1	Bolgan, M; Picciulin	EFFECTS OF	Is the Venice Lagoon Noisy? First Passive Listening Monitoring of the Ve	2016	PECO	E
22	282	10.1007/978-1-4939-2981-1	Roberts, L; Breithau	EFFECTS OF	Sensitivity of Crustaceans to Substrate-Borne Vibration	2016	PECO	E
23	286	10.1007/978-1-4939-2981-1	Sebastianutto, L; St	EFFECTS OF	Communicating the Issue of Underwater Noise Pollution: The Deaf as a	2016	PECO	P
24	287	10.1007/978-1-4939-2981-1	Sidorovskaia, NA; A	EFFECTS OF	Passive Acoustic Monitoring of the Environmental Impact of Oil Explorat	2016	PECO	E
25	304	10.1007/978-1-4939-2981-1	Willis, KL	EFFECTS OF	Underwater Hearing in Turtles	2016	PECO	E
26	306	10.1007/978-1-4939-2981-1	Zhang, XG; Guo, HG	EFFECTS OF	Noise-Dependent Fish Distribution in Kelp Beds	2016	PECO	E
27	309	10.1007/978-1-4939-2981-1	Lewandowski, J; Luc	EFFECTS OF	Summary Report Panel 3: Gap Analysis from the Perspective of Animal B	2016	PECO	O
28	311	10.1007/978-1-4939-2981-1	Aerts, LAM; Streever	EFFECTS OF	Modeled and Measured Underwater Sound Isopleths and Implications	2016	PECO	P
29	320	10.1007/978-1-4939-2981-1	Eggleston, DB; Lillis	EFFECTS OF	Soundscapes and Larval Settlement: Larval Bivalve Responses to Habita	2016	PECO	E
30	345	10.1007/978-1-4939-2981-1	Nehls, G; Rose, A; D	EFFECTS OF	Noise Mitigation During Pile Driving Efficiently Reduces Disturbance of	2016	PECO	E
31	361	10.1007/978-3-319-25658-1	Jennett, C; Cognetti	PARTICIPATC	Usability and Interaction Dimensions of Participatory Noise and Ecologi	2017	PECO	P
32	362	10.1007/978-3-319-27721-1	Roderick A. Suthers	VERTEBRATE	Vertebrate Sound Production and Acoustic Communication	2016	PECO	E
33	446	10.1007/BF00614503	ROMER, H; BAILEY, V	JOURNAL OF	INSECT HEARING IN THE FIELD. 3. MASKING BY NOISE	1989	PECO	E
34	454	10.1007/BF01651380	Bergen, F; Abs, M	JOURNAL FU	Etho-ecological study of the singing activity of the Blue Tit (<i>Parus caeruleus</i>)	1997	Language	-
35	463	10.1007/BF02242027	Kusters, E; van Rade	ZEITSCHRIFT	On the influence of military shooting ranges on the birds of the Wadder	1998	Language	-
36	470	10.1007/BF02465629	Rheindt, FE	JOURNAL FU	The impact of roads on birds: Does song frequency play a role in deter	2003	PECO	E
37	477	10.1007/BF03160888	Holland, CC; Honea	WETLANDS	Wetland degradation and loss in the rapidly urbanizing area of Portlan	1995	PECO	E
38	480	10.1007/s00018-015-1919-1	De Iriarte Rodriguez	CELLULAR AN	C-Raf deficiency leads to hearing loss and increased noise susceptibili	2015	PECO	P
39	532	10.1007/s00265-006-0188-8	Brumm, H; Slater, PJ	BEHAVIORAL	Ambient noise, motor fatigue, and serial redundancy in chaffinch song	2006	PECO	E
40	539	10.1007/s00265-013-1625-5	Penna, M; Zuniga, D	BEHAVIORAL	Strong responsiveness to noise interference in an anuran from the sout	2014	PECO	E
41	547	10.1007/s00267-001-0065-5	Forman, RTT; Reine	ENVIRONME	Road traffic and nearby grassland bird patterns in a suburbanizing land	2002	PECO	E
42	553	10.1007/s00267-010-0478-5	Smith, LA; Shaw, Fe	ENVIRONME	Impacts of Adjacent Land Use and Isolation on Marsh Bird Communities	2010	PECO	E
		README	CodeBook	List of all articles	List of excluded fulltexts			

Traceability of decisions : details of included studies

The list of included full-texts is mandatory

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
	ID_doc	Reference	Publication type	Lang.	Question address	Country	Region	GPS coordinate	Biological groups	LTI	LTI verge	Comparison	Study design	Outcomes	Susceptibility to bias	Narrative synthesis	Meta analysis
1	WOS_879	Anderson BS, Hunt JW, Phillips BM, Nicely PA, Vlaming V d., Connor V, et al. Integrated assessment of the impacts of agricultural drainwater in the Salinas River (California, USA). <i>Environ Pollut.</i> 2003;124:523-32.	Article	Eng.	Q1	United States of America	California, Salinas river	/	Benthic macroinvertebrates	Waterway	River sediments	- Upstream and downstream of the mouths of the agricultural drains	CI or CE	Abundance Species richness	Medium	Not included	Not included
2	WOS_1235	Armitage PD, Lattmann K, Kneebone N, Harris I. Bank profile and structure as determinants of macroinvertebrate assemblages - seasonal changes and management. <i>Regul Rivers Res Manag.</i> 2001;17:543-56.	Article	Eng.	Q1	United Kingdom	River Frome, Dorchester town	/	Macroinvertebrates	Waterway	River/water interface	- shallow vegetated bank - stepped bank - vertical earth bank - revetted bank	CI or CE	Abundance Species richness Community similarity	Medium	Not included	Not included
3	WOS_4888	Cavaillé P, Dommanget F, Daumergue N, Loucougaray G, Spiegelberger T, Tabacchi E, et al. Biodiversity assessment following a natural gradient of riverbank protection structures in French prealps rivers. <i>Ecol Eng.</i> 2013;53:23-30.	Article	Eng.	Q1	France	Rhône-Alpes region	See table 1	Carabidae (Coleoptera)	Waterway	Riverbank	- Vegetal embankment - Mixed embankment (vegetal and mineral) - Mineral embankment	CI or CE	Species richness	Medium	Included	Not included
4	WOS_5114	Chapman DS, Oxford GS, Dytham C. Process from pattern in the distribution of an endangered leaf beetle. <i>Ecography.</i> 2009;32:259-68.	Article	Eng.	Q1	United Kingdom	England, River Ouse, York city	/	<i>Chrysolina graminis</i> (Coleoptera)	Waterway	Riverbank	- Municipal management (sown grass managed by mowing) - Cattle-grazed - Sheep-grazed	CI or CE	Patch occupancy	Medium	Not included	Not included
5	WOS_8279	Dymitryszyn I. The effect of the construction and renovation of a highway bypass in Central Poland on the carabid beetle fauna (Coleoptera: Carabidae). <i>Eur J Entomol.</i> 2014;111:655-662.	Article	Eng.	Q1	Poland	Kujawsko-Pomorskie province, Skępe town, main road n° 10	/	Carabidae (Coleoptera)	Road	Roadside partly covered by plants	Before and after road renovation and roadsides reconstruction (widening of the road and its shoulders, replacement of the substratum and partial paving of the shoulders)	BACI	Abundance Species richness, Shannon index Community similarity Dominance, species traits	Low	Included	Not included
6	WOS_8853	Evrard M, Micha J-C. Relation entre la diversité du substrat et la diversité faunistique dans un bief belge de la rivière Meuse. <i>Ann Limnol - Int J Limnol.</i> 1995;31:93-103.	Article	Fr.	Q1	Belgium	Meuse river	/	Benthic macroinvertebrates	Waterway	Riverbank	- Natural substrates banks reinforcing - Artificial substrates banks reinforcing	CI or CE	Abundance Family richness, Shannon index	Medium	Not included	Not included
7	WOS_9124	Fell PE, Warren RS, Curtis AE, Steiner EM. Short-term Effects on Macroinvertebrates and Fishes of Herbiciding and Mowing Phragmites australis-dominated Tidal Marsh. <i>Northeast Nat.</i> 2006;13:191-212.	Article	Eng.	Q1	United States of America	Connecticut state, New London county, Lieutenant River	41°19'30"N, 72°	Macroinvertebrates	Waterway	Riparian marsh	- Untreated sites colonized with Phragmites - Treated sites: herbicide and mowing of <i>Phragmites australis</i>	CI or CE	Abundance Bray-Curtis similarity	Medium	Included	Not included
8	WOS_10923	Gollan JR, Reid CAM, Barnes PB, Wilkie L. The ratio of exotic-to-native dung beetles can indicate habitat quality in riparian restoration: Dung beetles in riparian restoration. <i>Insect Conserv Divers.</i> 2011;4:123-31.	Article	Eng.	Q1	Australia	Upper Hunter Valley	32°S, 151°E	Dung beetles (Coleoptera)	Waterway	Riparian habitat	- Unplanted (exotic herbs and pasture grasses) - 1-3 year old revegetated (trees and shrubs) riparian vegetation - 7-10 year old revegetated (trees and shrubs) riparian vegetation	CI or CE	Abundance Species richness Bray-Curtis similarity Ratio of the exotic-to-native beetles	Medium	Not included	Not included
9	WOS_10921	Gollan JR, Bruyn LL de, Reid N, Smith D, Wilkie L. Can ants be used as ecological indicators of restoration progress in dynamic environments? A case study in a revegetated riparian zone. <i>Ecol Indic.</i> 2011;11:1517-25.	Article	Eng.	Q1	Australia	New South Wales, Upper Hunter region	32°30'0"S, 151°	Ants (Hymenoptera)	Waterway	Riparian vegetation	- Riparian unplanted grassland - Riparian young revegetation - Riparian older revegetation	CI or CE	Abundance Species richness Bray-Curtis similarity, functional groups dissimilarity	Medium	Included	Not included
10	ZR_12929	Gonseth Y. Rhopalocères et structure du paysage. La faune des lépidoptères diurnes (Rhopalocera) des talus routiers et ferroviaires du Jura neuchâtelois. Université de Neuchâtel; 1996. http://doc.rero.ch/record/5267 .	PhD thesis	Fr.	Q1	Switzerland	Jura mountains of Neuchâtel	/	Butterflies (Lepidoptera)	Road and railway	Embankment	- Mowing - Grazing - No management	CI or CE	Species richness, Shannon index, Hill index Community composition, species traits	Medium	Not included	Not included
11	ZR_3357	Greenwood MT, Bickerton MA, Castella E, Large AR, Petts GE. The use of coleoptera (arthropoda: insecta) for floodplain characterization on the River Trent, UK. <i>Regul Rivers Res Manag.</i> 1991;6:321-332.	Article	Eng.	Q1	United Kingdom	England, River Trent	/	Carabidae and Staphylinidae (Coleoptera)	Waterway	Riparian habitat	Flood regulation	CI or CE	Abundance Community composition	Medium	Not included	Not included
12	ZR_3421	Grzybowska M. Development and habitat selection of chironomid	Article	Eng.	Q1	Poland	The River Widawka	Grabia: 52°31'	Macrobenthos	Waterway	Channel margins	- Stream with channel enlargement	CI or CE	Density	Medium	Not	Not

**A vous de jouer !
Soyez rigoureux et transparents !**



Faire des revues systématiques c'est n'avoir rien à cacher....





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Thank You



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